

PHYSICS

CLASS-XII

WAVE OPTICS

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VIBRANT ACADEMY
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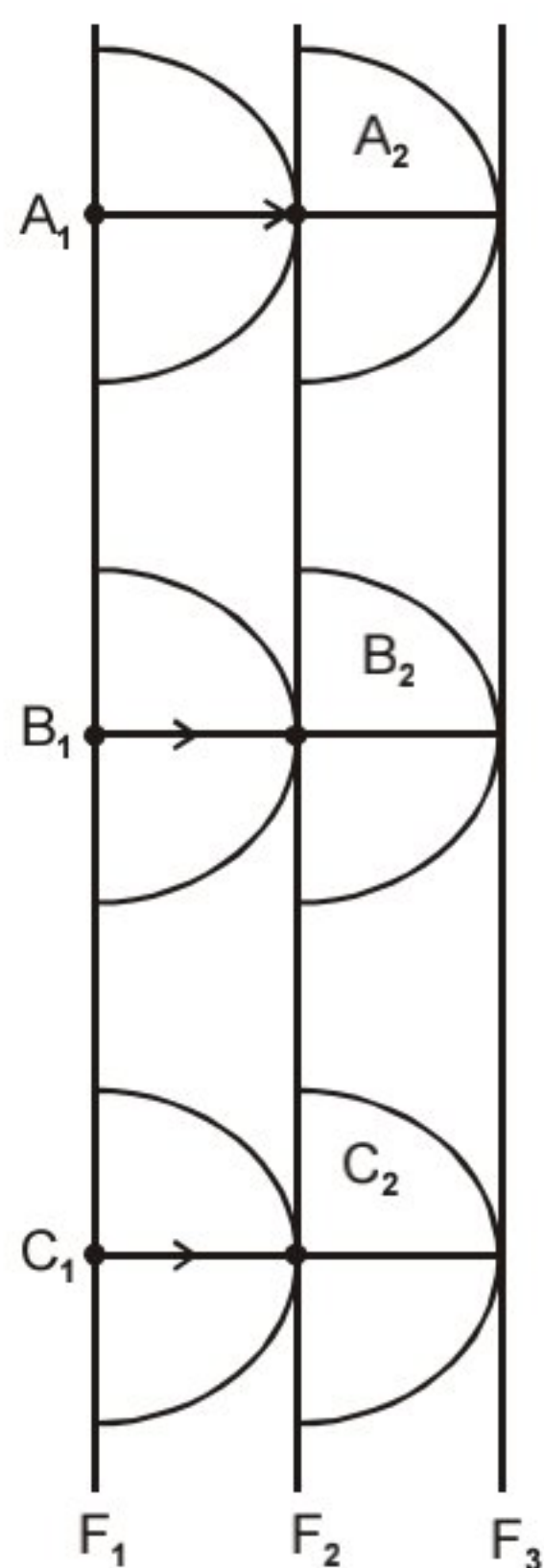
WAVE OPTICS

Introduction

The present chapter is concerned with *physical optics*, in which the wave of light of light is made manifest. In physical optics we consider what happens when light propagates around obstacles or through apertures that are *not* large compared to the wavelength. In such situations we encounter interference and diffraction. The distinction between interference and diffraction is not always clear-cut. Generally, interference refers to situations where waves from a small number of sources travel different paths and arrive at an observer with different phases. Diffraction is the spreading of waves after they travel around obstacles through apertures. According to Huygens's construction, *every point* on a wavefront is a source of wavelets.

HUYGEN'S CONSTRUCTION

Huygens, the Dutch physicist of the seventeenth century, gave a geometrical description of wave propagation. Huygen's idea is that we can think of every point on a wavefront as a new source of waves. According to Huygen's principle, what we observe is the result of adding up the waves from all these different sources. These are called secondary wave or wavelets. Huygen's principle is illustrated in figure in the simple case of a plane wave.



- (i) At time $t = 0$, we have a wavefront F_1 , F_1 separates those parts of the medium which are undisturbed from those where the wave has already reached.
- (ii) Each point on F_1 acts like a new source and sends out a spherical wave. After a time ' t ' each of these will have radius vt . These spheres are the secondary wavelets.
- (iii) After a time t , the disturbance would now have reached all points within the region covered by all these secondary waves. The boundary of this region is the new wavefront F_2 . Notice that F_2 is a surface tangent to all the spheres. It is called the forward envelope of these secondary wavelets.



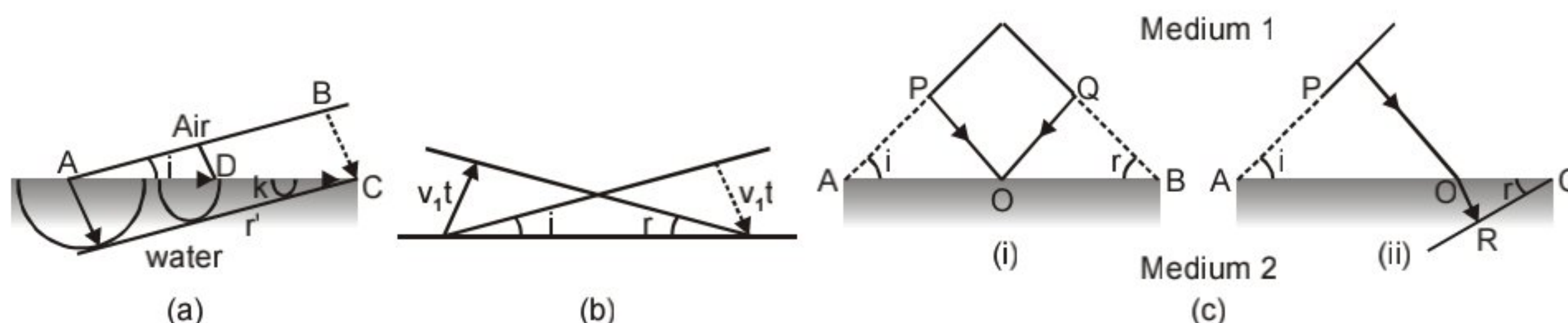
- (iv) The secondary wavelet from the point A₁ on F₁ touches F₂ at A₂. Draw the line connecting any point A₁ on F₁ to the corresponding point A₂ on F₂. According to Huygens A₁A₂ is a ray. It is perpendicular to the wavefronts F₁ and F₂ and has length vt. This implies that rays are perpendicular to wavefronts. Further, the time taken for light to travel between two wavefronts is the same along any ray. In our example, the speed v' of the wave has been taken to be the same at all points in the medium. In this case, we can say that the distance between two wavefronts is the same measured along any ray.
- (v) This geometrical construction can be repeated starting with F₂ to get the next wavefront F₃ a time t later, and so on. This is known as Huygen's construction.

Huygen's construction can be understood physically for wave in a material medium, like the surface of water. Each oscillating particle can set its neighbours into oscillation, and therefore acts as secondary source. But what if there is no medium, as for light travelling in vacuum? The mathematical theory, which cannot be given here, shows that the same geometrical construction works in this case as well.

Reflection and refraction

We can use a modified form of Huygen's construction to understand reflection and refraction of light. Figure (a) shows an incident wavefront which makes an angle 'i' with the surface separating two media, for example, air and water. The phase speeds in the two media are v₁ and v₂. We can see that when the point A on the incident wavefront strikes the surface, the point B still has to travel a distance BC = AC sin i, and this takes a time $t = BC / v_1 = AC (\sin i) / v_1$. After a time t, a secondary wavefront of radius v₂t with A as centre would have travelled into medium 2. The secondary wavefront with C as centre would have just started, i.e. would have zero radius. We also show a secondary wavelet originating from a point D in between A and C. Its radius is less than v₂t. The wavefront in medium 2 is thus a line passing through C and tangent to the circle centred on A. We can see that the angle r' made by this refracted wavefront with the surface is given by AE = v₂t = AC sin r'. Hence, $t = AC (\sin r') / v_2$. Equating the two expressions for 't' gives us the law of refraction in the form $\sin i / \sin r' = v_1 / v_2$. A similar picture is drawn in (fig. b) for the reflected wave which travels back into medium 1. In this case, we denote the angle made by the reflected wavefront with the surface by r, and we find that i = r. Notice that for both reflection and refraction, we use secondary wavelets starting at different times. Compare this with the earlier application (fig.) where we start them at the same time.

The preceding argument gives a good physical picture of how the refracted and reflected waves are built up from secondary wavelets. We can also understand the laws of reflection and refraction using the concept that the time taken by light to travel along different rays from one wavefront to another must be the same. (fig.) Shows the incident and reflected wavefronts when a parallel beam of light falls on a plane surface. One ray POQ is shown normal to both the reflected and incident wavefronts. The angle of incidence i and the angle of reflection r are defined as the angles made by the incident and reflected rays with the normal. As shown in figure (c), these are also the angles between the wavefront and the surface.



(fig.) (a) Huygen's construction for the (a) refracted wave. (b) Reflected wave. (c) Calculation of propagation time between wavefronts in (i) reflection and (ii) refraction.

We now calculate the total time to go from one wavefront to another along the rays. From (fig. (c)), we have, we have total time for light to reach from P to Q.

$$= \frac{PO}{v_1} + \frac{OQ}{v_1} = \frac{AO \sin i}{v_1} + \frac{OB \sin r}{v_1} = \frac{OA \sin i + (AB - OA) \sin r}{v_1} = \frac{AB \sin r + OA (\sin i - \sin r)}{v_1}$$

Different rays normal to the incident wavefront strike the surface at different point O and hence have different values of OA. Since the time should be the same for all the rays, the right side of equation must actually be independent of OA. The condition, for this to happen is that the coefficient of OA in eq. (should be zero, i.e., $\sin i = \sin r$). We, thus have the law of reflection, $i = r$. Figure also shows refraction at a plane surface separating medium 1 (speed of light v_1) from medium 2 (speed of light v_2). The incident and refracted wavefronts are shown, making angles i and r' with the boundary. Angle r' is called the angle of refraction. Rays perpendicular to these are also drawn. As before, let us calculate the time taken to travel between the two wavefronts along any ray.

$$\begin{aligned}\text{Time taken from P to R} &= \frac{PO}{v_1} + \frac{OR}{v_2} \\ &= \frac{OA \sin i}{v_1} + \frac{(AC - OA) \sin r'}{v_2} = \frac{AC \sin r'}{v_2} + OA \left(\frac{\sin i}{v_1} - \frac{\sin r'}{v_2} \right)\end{aligned}$$

This time should again be independent of which ray we consider. The coefficient of OA in equation is, therefore, zero.

$$\text{That is, } \frac{\sin i}{\sin r'} = \frac{v_1}{v_2} = n_{21}$$

where n_{21} is the refractive index of medium 2 with respect to medium 1. This is the Snell's law of refraction that we have already dealt with from eq. n_{21} is the ratio of speed of light in the first medium (v_1) to that in the second medium (v_2). Equation is, known as the Snell's law of refraction. If the first medium is vacuum,

$$\text{we have } \frac{\sin i}{\sin r'} = \frac{c}{v_2} = n_2$$

where n_2 is the refractive index of medium 2 with respect to vacuum, also called the absolute refractive index of the medium. A similar equation defines absolute refractive index n_1 of the first medium.

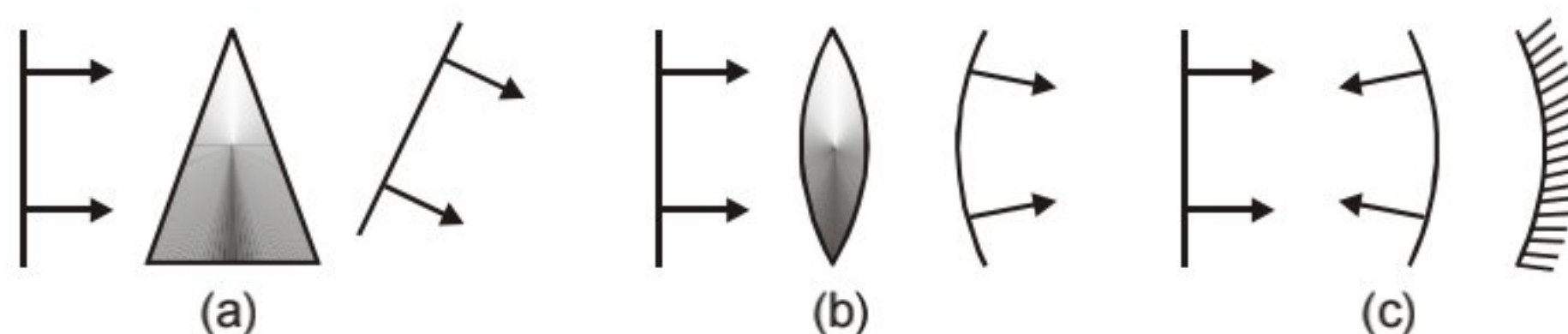
$$\text{From eq. we then get } n_{21} = \frac{v_1}{v_2} = \left(\frac{c}{n_1} \right) / \left(\frac{c}{n_2} \right) = \frac{n_2}{n_1}$$

The absolute refractive index of air is about 1.0003, quite close to 1. Hence, for all practical purposes, absolute refractive index of a medium may be taken with respect to air. For water, $n_1 = 1.33$, which means

$$v_1 = \frac{c}{1.33}, \text{ i.e. about 0.75 times the speed of light in vacuum. The measurement of the speed of light in}$$

water by Foucault (1850) confirmed this prediction of the wave theory.

Once we have the laws of reflection and refraction, the behaviour of prisms, lenses, and mirrors can be understood. These topics are discussed in details in the previous chapter. Here we just describe the behaviour of the wavefronts in these three cases (fig).



- (i) Consider a plane wave passing through a thin prism. Clearly, the portion of the incoming wavefront which travels through the greatest thickness of glass has been delayed the most. Since light travels more slowly in glass. This explains the tilt in the emerging wavefront.

- (ii) Similarly, the central part of an incident plane wave traverses the thickest portion of a convex lens and is delayed the most. The emerging wavefront has a depression at the centre. It is spherical and converges to a focus.
- (iii) A concave mirror produces a similar effect. The centre of the wavefront has to travel a greater distance before and after getting reflected, when compared to the edge. This again produces a converging spherical wavefront.
- (iv) Concave lenses and convex mirror can be understood from time delay arguments in a similar manner. One interesting property which is obvious from the pictures of wavefronts is that the total time taken from a point on the object to the corresponding point on the image is the same measured along any rays going through the centre are shorter. But because of the slower speed in glass, the time taken is the same as for rays travelling near the edge of the lens.

Diffraction

In the next section we shall discuss the experiment that first proved that light is a wave. To prepare for that discussion, we must introduce the idea of diffraction of waves, a phenomenon that we explore much more fully in later stage. Its essence is this: If a wave encounters a barrier that has an opening of dimensions similar to the wavelength, the part of the wave that passes through the opening will flare (spread) out -will *diffract-into* the region beyond the barrier. The flaring is consistent with the spreading of wavelets in the Huygens construction of Fig.. Diffraction occurs for waves of all types, not just light waves;

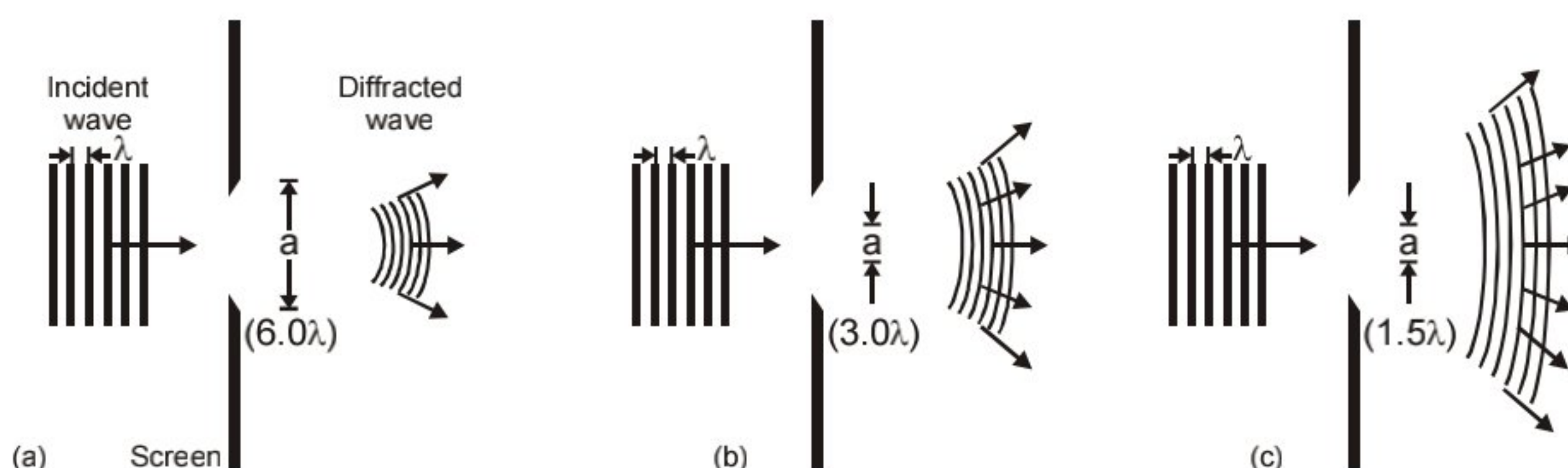


Fig. shows the diffraction of water waves traveling across the surface of water in a shallow tank. Figure shows the situation schematically for an incident plane wave of wavelength λ encountering a slit that has width $a = 6.0\lambda$ and extends into and out of the page. The wave flares out on the far side of the slit. Figures (with $a = 3.0\lambda$) and ($a = 1.5\lambda$) illustrate the main feature of diffraction: the narrower the slit, the greater the diffraction. Diffraction limits geometrical optics, in which we represent an electromagnetic wave with a ray. If we actually try to form a ray by sending light through a narrow slit, or through a series of narrow slits, diffraction will always defeat our effort because it always causes the light to spread. Indeed, the narrower we make the slits (in the hope of producing a narrower beam), the greater the spreading is. Thus, geometrical optics holds only when slits or other apertures that might be located in the path of light do not have dimensions comparable to or smaller than the wave length of the light.

Some points of Wave

- (i) In a wave motion, there is some quantity which changes its value with time and space. The quantity that changes with space and time, in terms of which the wave equation should be written, is the electric field existing in space where light travels.
- (ii) The light waves need no material medium to travel. They can propagate in vacuum.
- (iii) Because light waves are transverse, they can be polarized. If a plane wave of light is travelling along the x-direction, the electric field may be along the y-direction or along the z-direction or in any other direction in



the y-z plane. The equation of such a light wave may be written as

$$E = E_0 \sin \omega (t - x / v),$$

Where E_0 is the magnitude of the electric field at point x at time t.

- (iv) The speed of light in vacuum is always the same and is very nearly equal to 3×10^8 m/s.
- (v) The frequency of visible light varies from about 3800×10^{11} Hz to about 7800×10^{11} Hz. The corresponding wavelengths (obtained from $\lambda = c/v$) are 380 nm to 780 nm. Light of wavelength close to 780 nm appears red, and that close to 380 nm appears violet.
- (vi) Light of single wavelength is called monochromatic light.
- (vii) White light itself is a mixture of light of all wavelengths from about 380 nm to about 780 nm in appropriate proportion.

Coherent and Incoherent Sources

Why do we not commonly see interference effects with visible light? With light from a source such as the Sun, an incandescent bulb, or a fluorescent bulb, we do not see regions of constructive and destructive interference; rather, the *intensity* at any point is the sum of the intensities due to the individual waves. Light from anyone of these sources is, at the atomic level, emitted by a vast number of *independent* sources. Waves from independent sources are incoherent; they do not maintain a fixed phase relationship with each other. We cannot accurately predict the phase (for instance, whether the wave is at a maximum or at a zero) at one point given the phase at another point. Incoherent waves have *rapidly fluctuating* phase relationships. The result is an averaging out of interference effects, so that the total intensity (or power per unit area) is just the sum of the intensities of the individual waves. Only the superposition of coherent waves produces interference. Coherent waves must be locked in with a fixed phase relationship. *Coherent* and *incoherent* waves are idealized extremes; all real waves fall somewhere between the extremes. The light emitted by a laser can be highly coherent—two points in the beam can be coherent even if separated by as much as several kilometers. Light from a distant point source (such as a star other than the Sun) has some degree of coherence.

Thomas Young (1773-1829) performed the first visible-light interference experiments using a clever technique to obtain two coherent light sources from a single source. When a single narrow slit is illuminated, the light wave that passes through the slit diffracts or spreads out. The single slit acts as a single coherent source to illuminate two other slits. These two other slits then act as sources of coherent light for interference. Young's technique for illuminating two slits with coherent light. The single slit on the left serves as a source of coherent light.

The "snapshot" of fig. depicts the interference of the overlapping waves. However, we cannot see evidence for the interference except where a viewing screen C intercepts the light. Where it does so, points of interference maxima form visible bright rows—called *bright bands*, *bright fringes*, or (loosely speaking) *maxima* that extend across the screen (into and out of the page in Fig.).

Dark regions

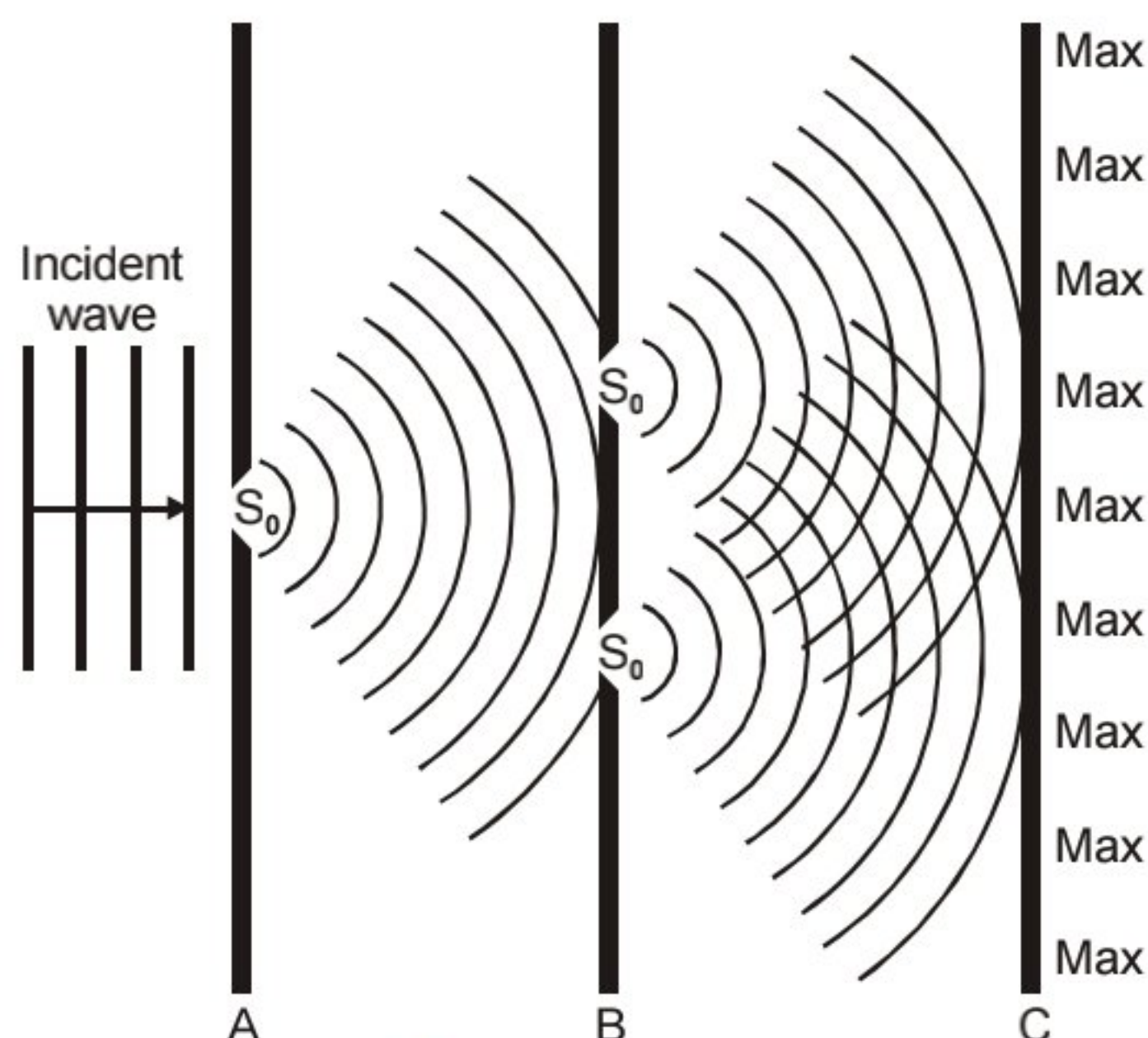


Fig. in Young's interference experiment, incident monochromatic light is diffracted by slit S_0 , which then acts as a point source of light that emits semicircular wavefronts. As that light reaches screen B , it is diffracted by slits S_1 and S_2 , which then act as two point sources of light. The light waves traveling from slits S_1 and S_2 overlap and undergo interference, forming an interference pattern of maxima and minima on viewing screen C . This figure is a cross section; the screens, slits, and interference pattern extend into and out of the page. Between screens B and C , the semicircular wavefronts centered on S_2 depict the waves that would be there if only S_2 were open. Similarly, those centered on S_1 depict waves that would be there if only S_1 were open.

INTENSITY OF TWO SOURCE INTERFERENCE

We now obtain an expression for the distribution of intensity of two coherent sources that are in phase. The wave function in this case is the electric field. We assume that the slits are narrow enough for diffraction to spread light from each slit uniformly over the screen. Thus, the amplitude of the fields at any point on the screen will be equal. At a given point of the screen the fields due to S_1 and S_2 are

$$E_1 = E_0 \sin(\omega t); \quad E_2 = E_0 \sin(\omega t + \phi)$$

where the phase difference ϕ depends on the path difference $\delta = r_2 - r_1$. Since one wavelength λ corresponding to a phase change of 2π , a distance δ corresponds to a phase change ϕ given by $\phi / 2\pi = \delta / \lambda$. If the screen is far from the slits, $\delta \approx d \sin \theta$ (see the ref. figure for YDSE), therefore

$$\phi = \frac{2\pi\delta}{\lambda} = \frac{2\pi d \sin \theta}{\lambda}$$

The resultant field is found from the principle of superposition :

$$E = E_1 + E_2 = E_0 \sin(\omega t) + E_0 \sin(\omega t + \phi)$$

By using the trigonometric identity $\sin A + \sin B = 2 \sin [(A + B)/2] \cos [(A - B)/2]$, we obtain

$$E = 2E_0 \cos\left(\frac{\phi}{2}\right) \sin\left(\omega t + \frac{\phi}{2}\right)$$

The amplitude of the resultant wave is $2E_0 \cos(\phi/2)$. The intensity of a wave is proportional to the square of the amplitude, so from equation we have

$$I = 4I_0 \cos^2 \cos^2\left(\frac{\phi}{2}\right)$$

where $I_0 \propto E_0^2$ is the intensity due to a single source. This function is plotted in fig. of pt (iv). The maxima occur when $\phi = 0, 2\pi, 4\pi, \dots = 2m\pi$. At these points $I = 4I_0$; that is the intensity is four times that of a single source. The minima ($I = 0$) occur when $\phi = \pi, 3\pi, 5\pi, \dots = (2m + 1)\pi$.

Fig.(a) shows the waves emitted by sources S_1 and S_2 . The waves from the source start in phase and arrive in phase, leading to constructive interference at the point.

In fig. (a) the distances travelled by waves differ by any integer number of wavelengths.

$$x_2 - x_1 = \lambda, 2\lambda, 3\lambda, \dots, n\lambda.$$

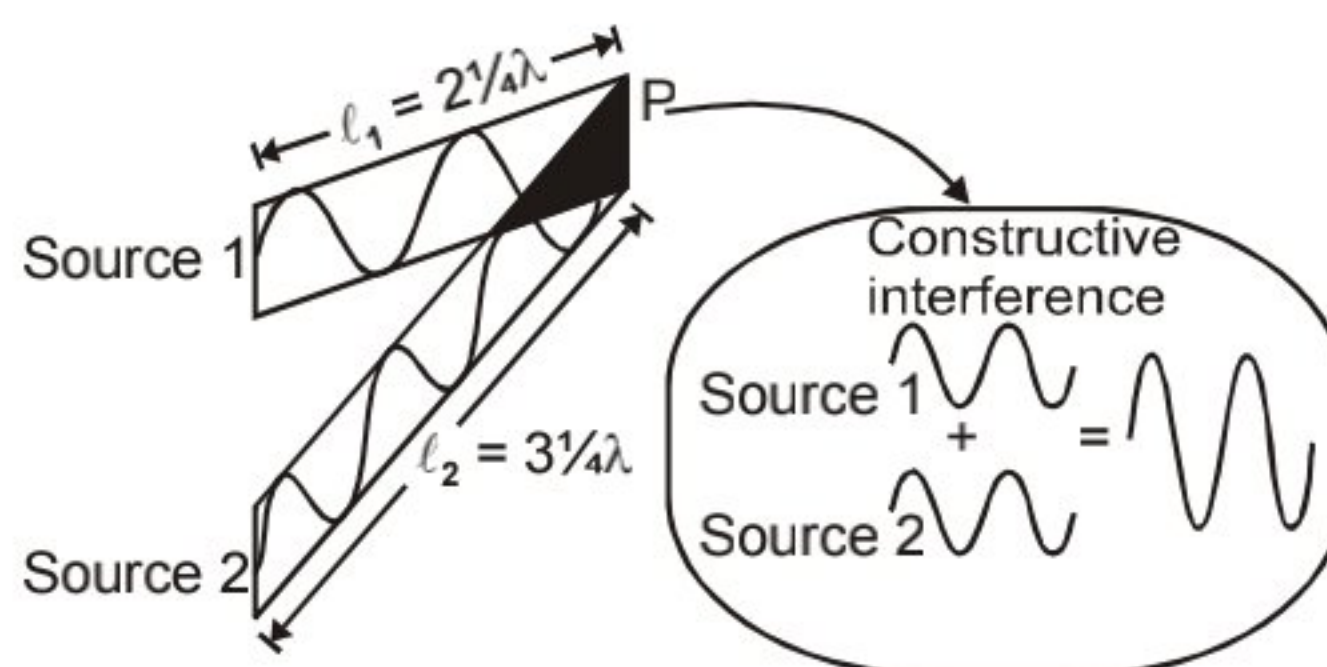
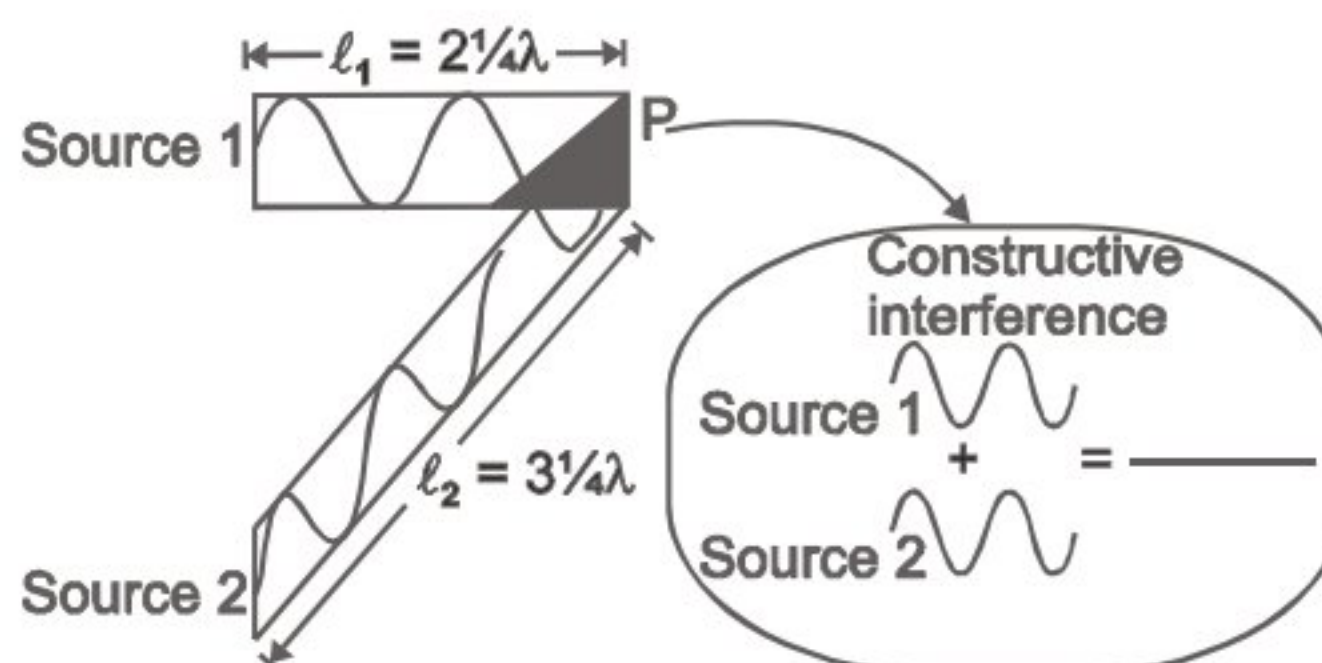


Fig. (a) shows two waves starting in phase but arriving in opposite phase.

In fig. (b) the distance travelled by waves differ by odd integer number of wavelengths.

$$x_2 - x_1 = \frac{\lambda}{2}, \frac{3}{2}\lambda, \frac{5}{2}\lambda, \dots \left(n - \frac{1}{2}\right)\lambda$$



- (i) If amplitudes of waves arriving at point P on the screen are different then resultant intensity is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

Also, $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$, when $\cos \delta = 1$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2, \text{ when } \cos \delta = -1$$

(ii)
$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 = \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2 = \left(\frac{r+1}{r-1} \right)^2$$

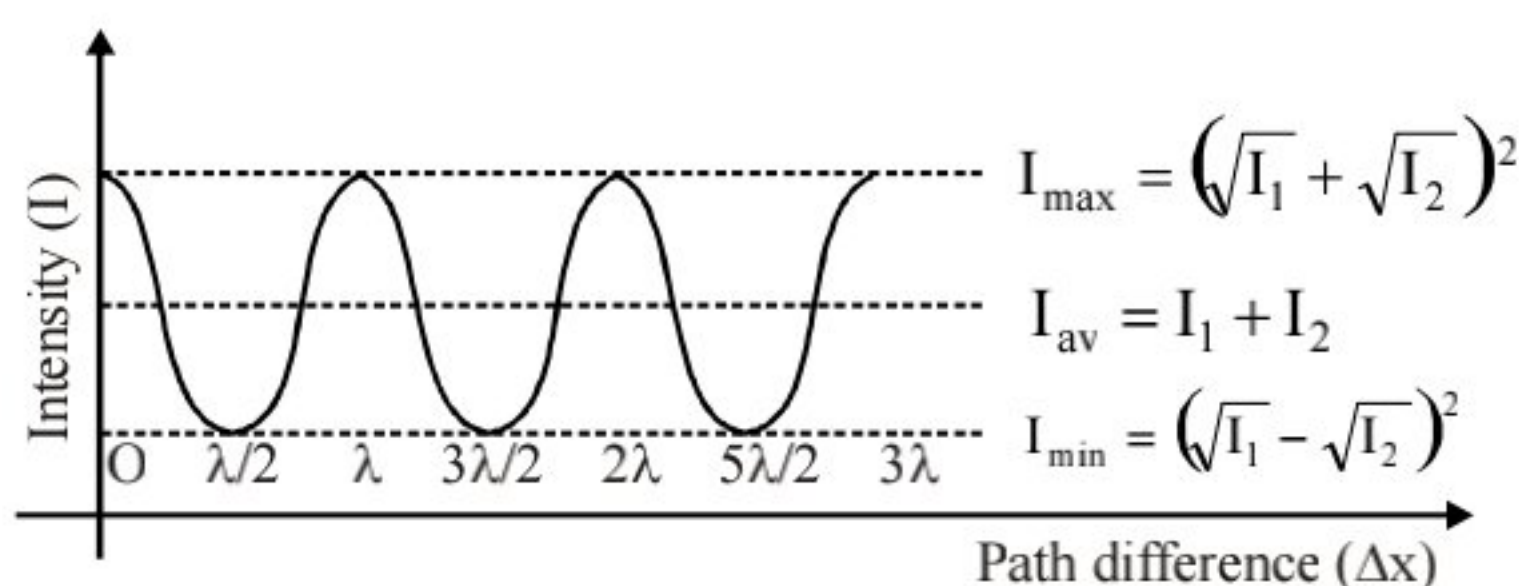
where $r = \frac{A_1}{A_2} = \sqrt{\frac{I_1}{I_2}}$

- (iii) The phenomenon of interference is based on conservation of energy. There is no destruction of energy in the interference phenomenon. The energy which apparently disappears at the minima, has actually been transferred to the maxima where the intensity is greater than that produced by the two beams acting separately.

$$I_{\text{av}} = \frac{\int_0^{2\pi} I d\delta}{\int_0^{2\pi} d\delta} = \frac{1}{2\pi} \int_0^{2\pi} (I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta) d\delta = I_1 + I_2 \quad \therefore \int_0^{2\pi} \cos \delta d\delta = 0$$

as the average value of intensity is equal to the sum of individual intensities, therefore the energy is not destroyed but merely redistributed in the interference pattern.

- (iv) All maxima are equally spaced and equally bright. This is true for minima as well. Also interference maxima and minima are alternate. The intensity distribution in interference pattern is shown in figure.



(iv) Path difference (Δx) and phase difference (ϕ) are related as given below :

$$\lambda \text{ path difference} = 2\pi \text{ phase difference}$$

$$\text{or } \Delta x = \frac{\lambda}{2\pi} \phi$$

Coherent source :

The sources which produce sustained, i.e. observable interference are called coherent sources. In case of interference as $I = [I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi]$, interference will be sustained if the phase difference ϕ at a given point does not vary with time. If the interfering wave are :

$$y_1 = A_1 \sin (\omega_1 t - k_1 x_1 + \phi) \text{ and } y_2 = A_2 \sin (\omega_2 t - k_2 x_2 + \phi_2)$$

$$\phi = (\omega_1 - \omega_2) t + (k_2 x_2 - k_1 x_1) + (\phi_1 - \phi_2)$$

So ϕ will not vary with time if :

- (a) $(\phi_1 - \phi_2) = \phi_0$ is constant, i.e., the initial phase difference between the wave does not vary with time and
- (b) $(\omega_1 - \omega_2) t = 0$, i.e., $f_1 = f_2$. But for a wave as $v = f\lambda$, $f_1 = f_2$ will also mean $\lambda_1 = \lambda_2$, i.e., $k_1 = k_2$ [as $k = 2\pi / \lambda$].

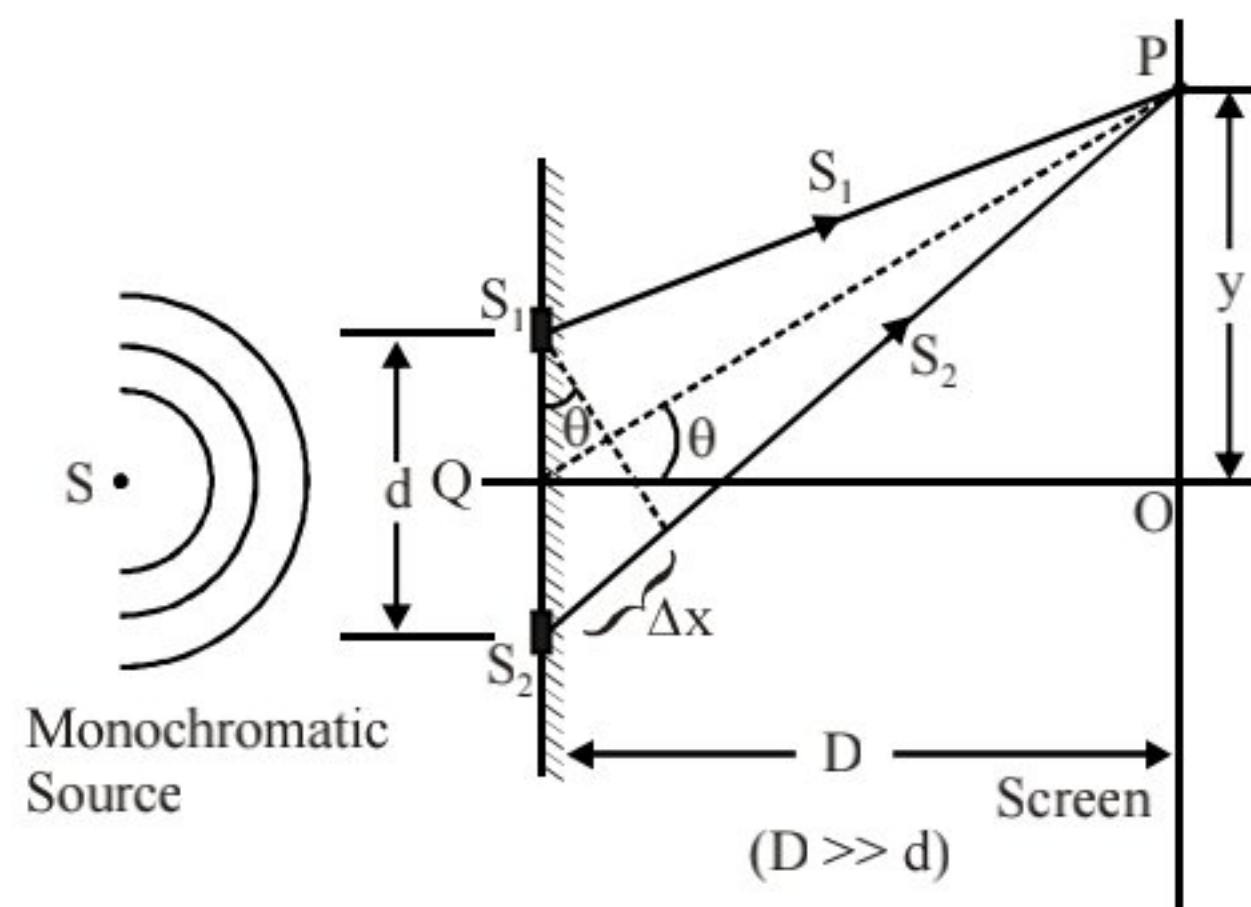
i.e., the two wave are of same frequency and wavelength. So two sources will be coherent if and only if they produce wave of same frequency (and hence wavelength) and have a constant initial phase difference. So in case of two coherent sources.

$$\phi = \frac{2\pi}{\lambda} (\Delta x) + \phi_0 \text{ with } \phi_0 = (\phi_1 - \phi_2)$$

Now as in general emission of light from atoms is random, rapid and independent of each other, ϕ_0 cannot remain constant with time and hence two independent light source cannot be coherent.

YOUNG'S DOUBLE SLIT EXPERIMENT (YDSE)

The experiment set up for Young's double slit experiment is shown in figure. Light after passing through a pin hole 'S' is allowed to fall on thin slits ' S_1 ' and ' S_2 ' placed symmetrically w.r.t. 'S'. A screen is placed at a distance 'D' from S_1 and S_2 .



Geometric construction for describing Young's double-slit experiment

Let 'P' be the point, at which we want to investigate the intensity. Two rays S_1P and S_2P starting from S_1 and S_2 reach P and interfere with each other.

If Δx is the path difference between two rays,

$$\Delta x = S_2P - S_1P \approx d \sin \theta \approx \frac{dy}{D} \quad \left(\because \text{For small angle } \sin \theta \approx \tan \theta = \frac{y}{D} \right)$$



This equation assumes that S_1P and S_2P are parallel, which is approximately true because D is much greater than d .

- (a) **Maxima** : Point 'P' will be a bright spot if the path difference Δx is integral multiple of λ .

$$y_n = \frac{nD\lambda}{d} \quad \text{where, } n = 0, 1, 2, 3, \dots$$

Thus, bright spots are obtained at distances, $0, \frac{\lambda D}{d}, \frac{2\lambda D}{d}, \frac{3\lambda D}{d}, \dots$ from O.

- (b) **Minima** : Point 'P' will be a dark spot if the path difference ' Δx ' is an odd multiple of $\frac{\lambda}{2}$.

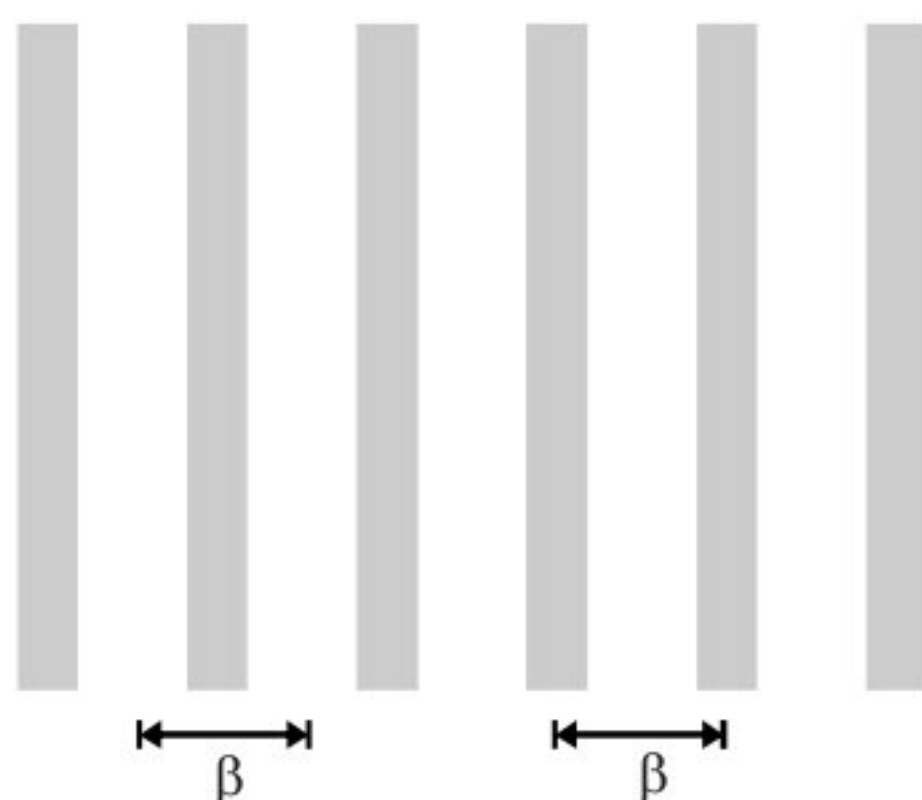
i.e., if
$$\frac{yd}{D} = \frac{(2n+1)\lambda}{2}$$

$$\therefore y_n = \frac{(2n+1)\lambda D}{2d} \quad \text{where, } n = 0, 1, 2, 3, \dots$$

Thus, dark spots are obtained at distances, $\frac{\lambda D}{2d}, \frac{3\lambda D}{2d}, \frac{5\lambda D}{2d}, \dots$ from O.

- (c) **Fringe width (β)**

It is the distance between two consecutive bright or dark fringes.



Let y_n and y_{n-1} respectively, be the distances of n^{th} and $(n-1)^{\text{th}}$ bright fringe from O,

$$\beta = [y_n - y_{n-1}] = n \frac{\lambda D}{d} - (n-1) \frac{\lambda D}{d} = \frac{\lambda D}{d} (n - n + 1)$$

or
$$\beta = \frac{\lambda D}{d}$$

Similarly, it can be proved that distance between two consecutive dark fringes, β is given by

$$\beta = \frac{\lambda D}{d}$$

$$\therefore \beta = \beta = \lambda D / d$$

Hence, the bright and dark fringes are equally spaced.

Remarks :

- (i) If whole apparatus is immersed in liquid of refractive index μ then,

$$\beta = \frac{\lambda D}{\mu d} \text{ i.e., fringes width decreases}$$

- (ii) Some times in numerical problems, angular fringe width (ω) is given which is defined as angular separation between two consecutive maxima or minima

$$\omega = \frac{\beta}{D} = \frac{\lambda}{d}$$

In medium, other than air or vacuum,

$$\omega = \frac{\lambda}{\mu d}$$

- (iii) $\Delta x = \frac{y d}{D}$ is valid when angular position of maxima or minima is less than $\frac{\pi}{6}$. However $\Delta x = d \sin \theta$ is valid for larger values of θ provided $d \ll D$.

- (iv) Central bright fringe (CBF) is a point on screen where path difference is zero. In above case CBF is formed at O. But in many situation it may not be located symmetrically w.r.t. slits.

- (v) If white light is used instead of monochromatic light then, interference pattern consists of white central bright fringe surrounded by few coloured fringes and then uniform illumination due to overlapping of interference pattern on each wavelength.

- (vi) If the interference experiment is performed with bichromatic light, the bright fringes of two wavelength will be coincident for the first time under following condition.

$$Y = n (\beta)_{\text{Longer}} = (n + 1) \beta_{\text{Shorter}} \text{ or } n \lambda_{\text{Longer}} = (n + 1) \lambda_{\text{Shorter}}$$

- (vii) In many numerical problems we have to calculate number of maxima or minima. We know that for maximum.

$$\sin \theta = \frac{n \lambda}{d} \quad \text{or} \quad n = \frac{d \sin \theta}{\lambda}$$

$$\therefore n \frac{d}{\lambda} \neq 1 \quad (\because \sin \theta \leq 1)$$

$$n_{\text{highest}} = \left[\frac{d}{\lambda} \right]$$

Lighest order of maxima on one side.

Suppose in some question $\frac{d}{\lambda}$ works out to be 2.3 so, permissible values of n are 0, ± 1 , ± 2 . Hence, total

5 maxima will be obtained on screen.



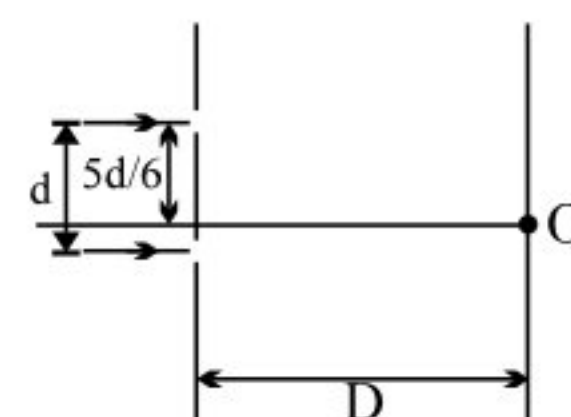
Ex. In YDSE experiment the distance between slits is $d = 0.25 \text{ cm}$ and the distance of screen $D = 120 \text{ cm}$ from slits. If the wavelength of light used is $\lambda = 6000 \text{ \AA}$ and I_0 is the intensity of central maximum, at what distance from the centre of the intensity will be $\frac{I_0}{2}$?

- (A) $4.8 \times 10^{-5} \text{ m}$
(C) $3.6 \times 10^{-5} \text{ m}$

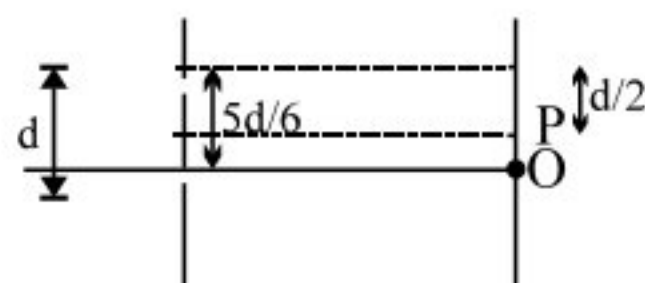
- (B*) $7.2 \times 10^{-5} \text{ m}$
(D) $5.4 \times 10^{-5} \text{ m}$

Ex. In a YDSE experiment, I_0 is given to be the intensity of the central bright fringe and β is the fringe width. Then, at a distance y from Central Bright Fringe, the intensity will be
(A) $I_0 \cos(\pi y/\beta)$
(C) $I_0 \cos(2\pi y/\beta)$
(B*) $I_0 \cos^2(\pi y/\beta)$
(D) $I_0 \cos^2(\pi y/2\beta)$

Ex. In the figure, if a parallel beam of white light is incident on the plane of the slits then the distance of the white spot on the screen from O is [assume $d \ll D$, $\lambda \ll d$]
(A) 0
(B) $d/2$
(C*) $d/3$
(D) $d/6$



Sol. White spot will be central max., where path diff. is zero



$$OP = 5d/6 - d/2 = d/3 \quad \text{Ans.}$$

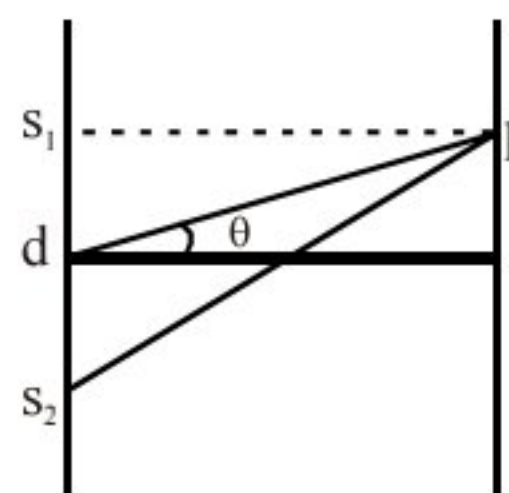
Ex. In a Young's double slit experiment, the separation between the slits is d , distance between the slit and screen is D ($D \gg d$). In the interference pattern, there is a maxima exactly in front for each slit. Then the possible wavelength(s) used in the experiment are
(A) $d^2/D, d^2/2D, d^2/3D$
(C*) $d^2/2D, d^2/4D, d^2/6D$
(B) $d^2/D, d^2/3D, d^2/5D$
(D) none of these

Sol. $s_2p - s_1p = \frac{d \times y}{D} = \frac{d \times (d/2)}{D} = \frac{d^2}{2D}$

$$\frac{d^2}{2D} = n\lambda$$

$$\lambda = \frac{d^2}{2nD}, n = 1, 2, \dots$$

$$\lambda = \frac{d^2}{2D}, \frac{d^2}{4D}, \frac{d^2}{6D}$$



- Ex.1** The two coherent sources of intensity ratio 2 : 8 produce an interference pattern. The values of maximum and minimum intensities will be respectively
 (A) I_1 and $9I_1$ (B) $9I_1$ and I_1 (C) $2I_1$ and $8I_1$ (D) $8I_1$ and $2I_1$
 Where I_1 is the intensity of first source

Sol. $I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$ (1)

According to question

$$\frac{I_1}{I_2} + \frac{2}{8} = \frac{1}{4}$$

$$\therefore I_2 = 4I_1$$

From eqs. (1) and (2)

$$I_{\max} = I_1 + 4I_1 + 2\sqrt{4I_1^2}$$

$$= 5I_1 + 4I_1$$

$$I_{\max} = 9I_1$$

$$I_{\min} = I_1 + I_2 - 2\sqrt{4I_1^2}$$

from eqs. (2) and (4)

$$I_{\min} = I_1 + 4I_1 - 2\sqrt{4I_1^2}$$

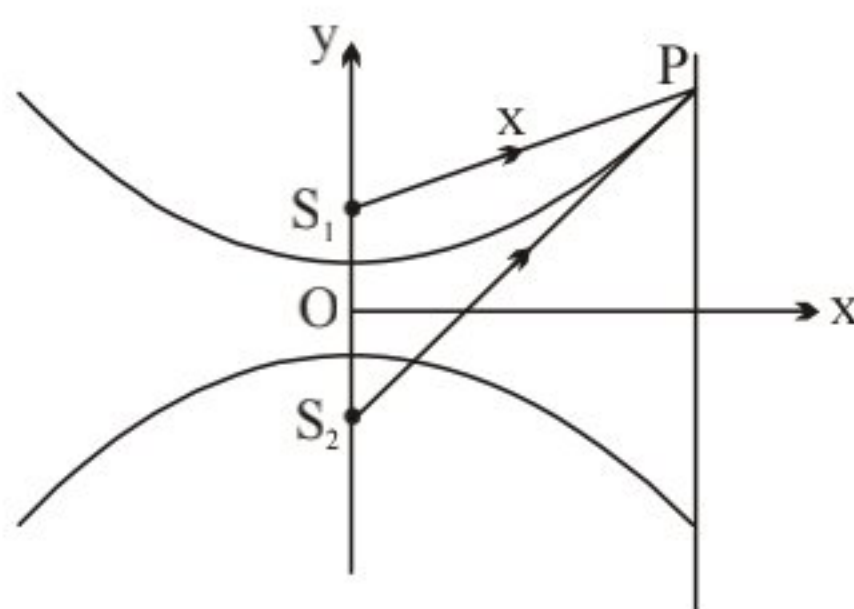
$$I_{\min} = I_1$$

Hence the correct answer will be (2)

SHAPE OF INTERFERENCE FRINGES IN YDSE

We discuss the shape of fringes when two pinholes are used instead of the two slits in YDSE.

Fringes are locus of points which move in such a way that its path difference from the two slits remains constant.



$$S_2P - S_1P = D = \text{constant} \quad \text{.....(1)}$$

If $\Delta = \pm \frac{\lambda}{2}$, the fringe represents 1st minima.

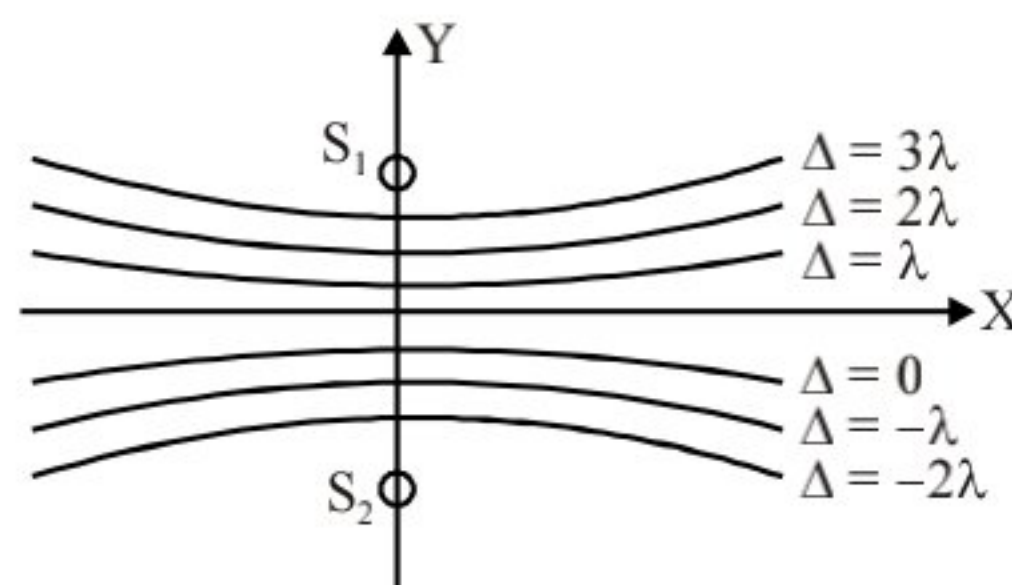
If $\Delta = \pm \frac{3\lambda}{2}$ it represents 2nd minima

If $\Delta = 0$ it represents central maxima,

If $\Delta = \pm \lambda$, it represents 1st maxima etc.

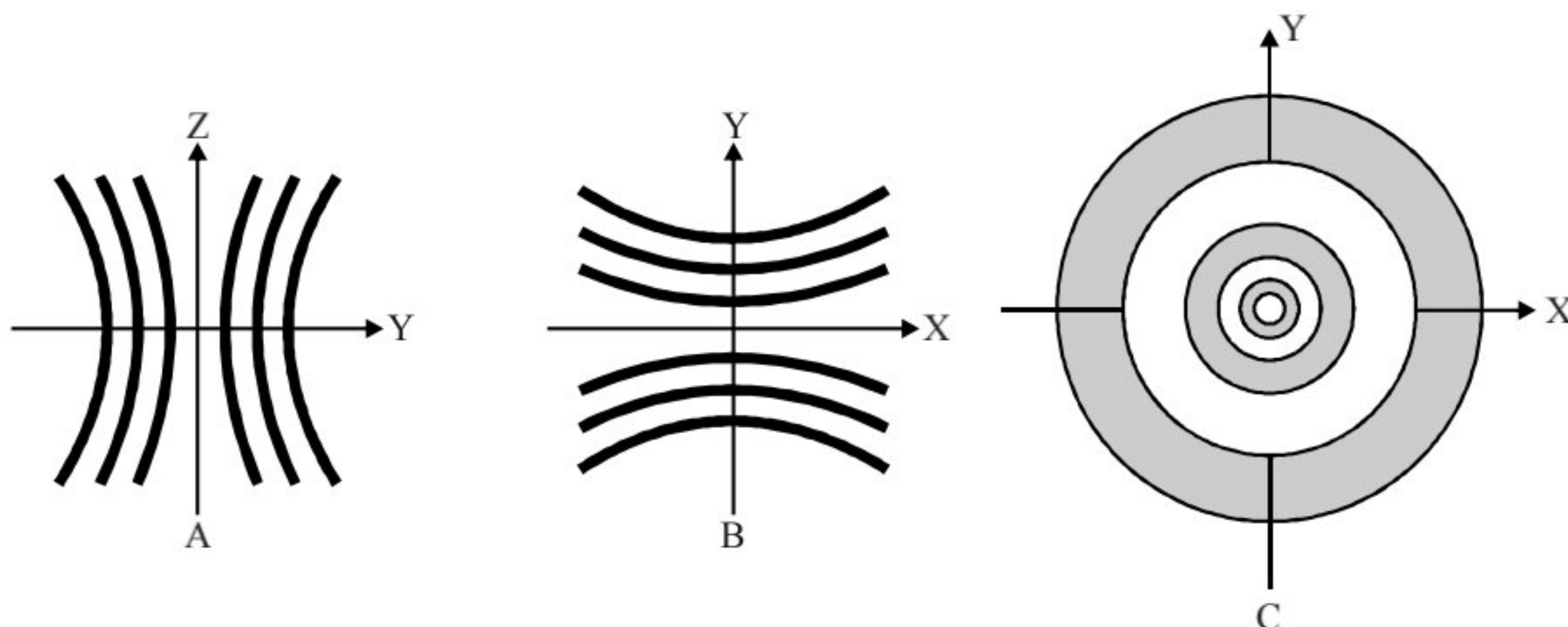
Equation (1) represents a hyperbola with its two foci at S_1 and S_2

The interference pattern which we get on screen is the section of hyperboloid of revolution when we revolve the hyperbola about the axis S_1S_2 .

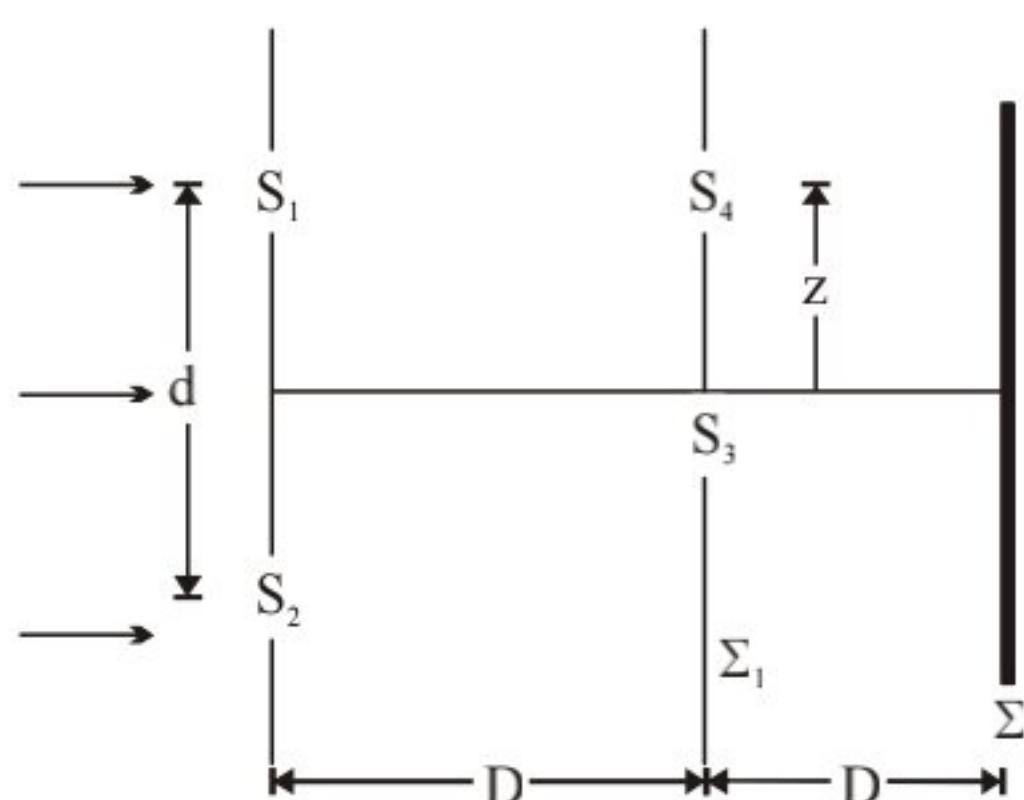


- (A) If the screen is $\perp$ to the X-axis, i.e. in the YZ plane, as is generally the case, fringes are hyperbolic with a straight central section.
- (B) If the screen is in the XY plane, again fringes are hyperbolic.
- (C) If screen is $\perp$ to Y-axis (along S_1S_2) i.e. in the XZ plane, fringes are concentric circles with center on the axis

S_1S_2 ; the central fringe is bright if $S_1S_2 = n\lambda$ and dark if $S_1S_2 = (2n - 1) \frac{\lambda}{2}$.



Ex. Consider the situation shown in figure. The two slits S_1 and S_2 placed symmetrically around the central line are illuminated by a monochromatic light of wavelength λ . The separation between the slits is d . The light transmitted by the slits falls on a screen Σ_1 placed at a distance D from the slits. The slit S_3 is at the central line and the slits S_4 is at a distance z from S_3 . Another screen Σ_2 is placed a further distance D away from Σ_1 . Find the ratio of the maximum to minimum intensity observed on Σ_2 if z is equal to



- (a) $\frac{D\lambda}{2d}$ (b) $\frac{\lambda D}{4d}$

Sol. (a) Let I is intensity due to slits S_1 & S_2 on screen Σ_1 . Further, intensity at any point on screen Σ_1 is given by

$$I_p = 4I \cos^2 \left(\frac{\phi}{2} \right)$$

At slit S_3 , $\phi = 0$

$$\therefore I_{S_3} = 4I$$

At slit S_4 ,
$$\Delta x = \frac{dz}{D} = \frac{\lambda}{2}$$

$\therefore \phi = \pi \Rightarrow I_{S_4} = 0$

Now on screen Σ_2
$$I_{\max} = (\sqrt{I_{S_3}} + \sqrt{I_{S_4}})^2 = 4I$$

$$I_{\min} = (\sqrt{I_{S_3}} - \sqrt{I_{S_4}})^2 = 4I$$

$\therefore \frac{I_{\max}}{I_{\min}} = 1 \text{ Ans}$

(b)
$$z = \frac{D\lambda}{4d}$$

$$I_{S_3} = 4I$$

At slit S_4 ,
$$\Delta x = \frac{dz}{D} = \frac{\lambda}{4}$$

$\therefore \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

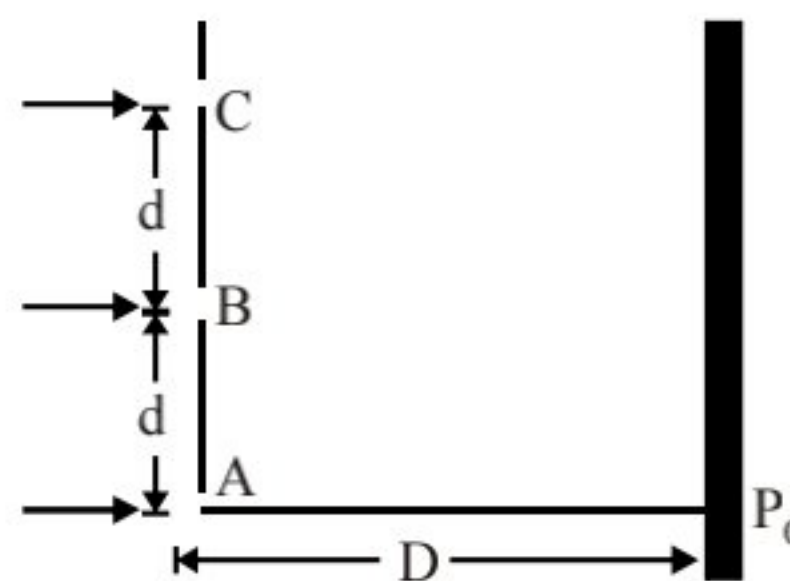
$\therefore I_{S_4} = 4I \cos^2\left(\frac{\pi}{4}\right) = 2I$

$\therefore I_{\max} = (\sqrt{I_{S_3}} + \sqrt{I_{S_4}})^2 = (\sqrt{4I} + \sqrt{2I})^2$
 $= I(2 + \sqrt{2})^2$

Similarly,
$$I_{\min} = (\sqrt{I_{S_3}} - \sqrt{I_{S_4}})^2 = (\sqrt{4I} - \sqrt{2I})^2 = I(2 - \sqrt{2})^2$$

$\therefore \frac{I_{\max}}{I_{\min}} = \left(\frac{2 + \sqrt{2}}{2 - \sqrt{2}}\right)^2 \text{ Ans.}$

Ex. Figure shows three equidistant slits being illuminated by a monochromatic parallel beam of light. Let $BP_0 - AP_0 = \lambda/3$ and $D \gg \lambda$. (a) Show that in this case $d = \sqrt{2\lambda D/3}$. (b) Show that the intensity at P_0 is three times the intensity due to any of the three slits individually.

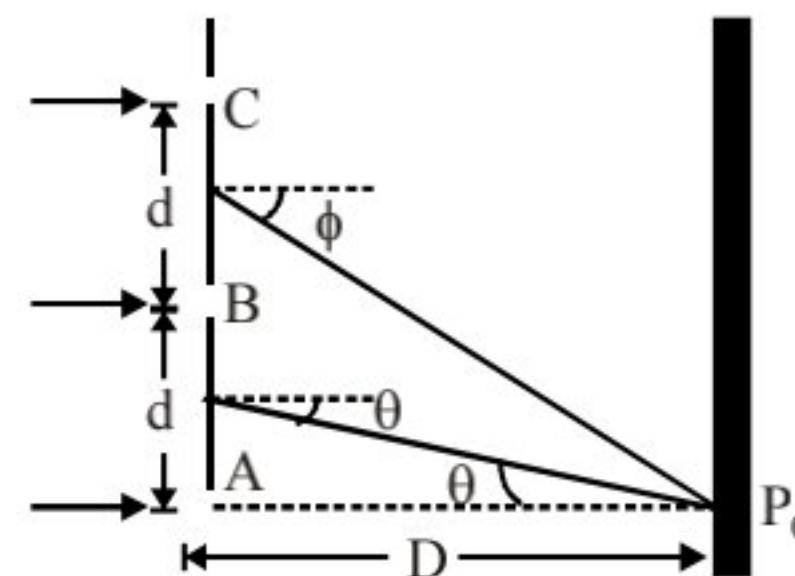


Sol. $BP_0 - AP_0 = \frac{\lambda}{3}$

or $d \sin \theta = \frac{\lambda}{3}$

or $d \tan \theta = \frac{\lambda}{3}$ (For small angle $\tan \theta \approx \sin \theta \approx \theta$)

$\therefore d \left(\frac{d/2}{D} \right) = \frac{\lambda}{3}$ or $d = \sqrt{\frac{2F\lambda}{3}}$



(b) Δx_{AB} = path difference between waves coming from A and B = $\frac{\lambda}{3}$

$\therefore \phi_{AB}$ = phase difference

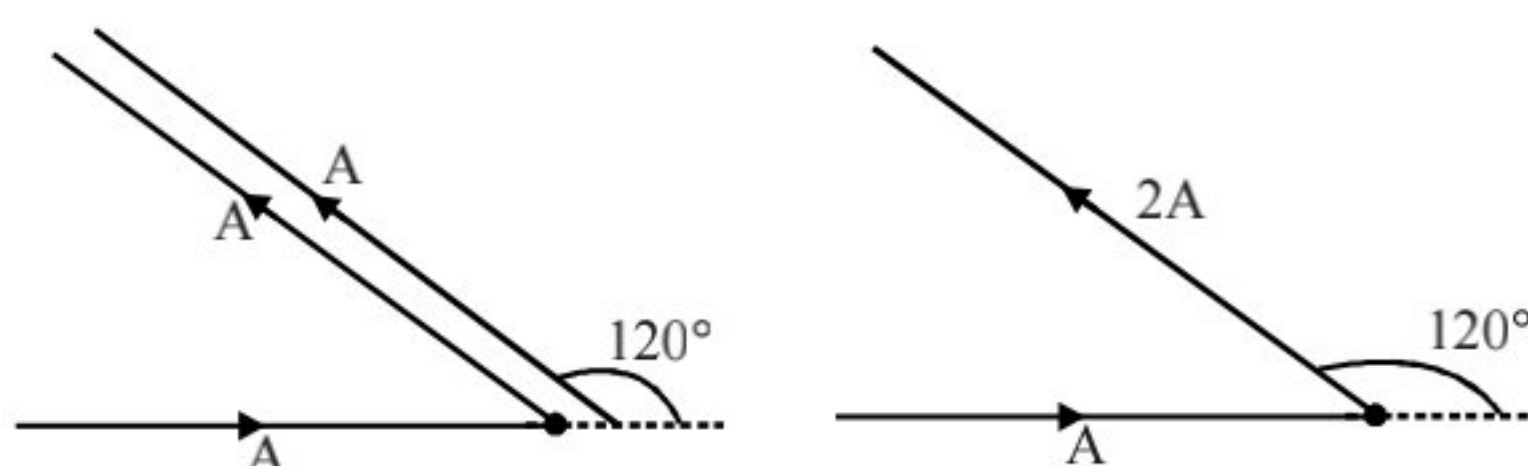
$= \frac{2\pi}{\lambda} \Delta x_{AB} = \frac{2\pi}{3}$

Similarly, $\Delta x_{BC} = d \sin \phi$

$= d \left(\frac{3d/2}{D} \right) = \frac{3d^2}{2D} = \lambda$

$\therefore \phi_{BC} = 2\pi$

Now, phase diagram of the waves arriving at P_0 is as shown below :



$\therefore$ Amplitude of resultant wave is given by

$A' = \sqrt{A^2 + (2A)^2 + 2(A)(2A)\cos 120^\circ} = \sqrt{3} A$

As intensity $(I) \propto A^2$

$\therefore$ Intensity at P_0 will be three times the intensity due to any of the three slits individually.

YDSE WITH WHITE LIGHT :

The central maxima will be white because all wavelengths will constructively interfere here. However slightly below (or above) the position of central maxima fringes will be coloured for example if P is a point on the screen such that

$S_2P - S_1P = \frac{\lambda_{\text{violet}}}{2} = 190 \text{ nm},$

completely destructive interference will occur for violet light. Hence we will have a line devoid of violet colour that will appear reddish. And if



$$S_2P - S_1P = \frac{\lambda_{\text{red}}}{2} \simeq 350 \text{ nm},$$

completely destructive interference for red light results and the line at this position will be violet. The coloured fringes disappear at points far away from the central white fringe ; for these points there are so many wavelengths which interfere constructively, that we obtain a uniform white illumination for example if

$$S_2P - S_1P = 3000 \text{ nm},$$

then constructive interference will occur for wavelengths $\lambda = \frac{3000}{n} \text{ nm}$. In the visible region these wavelengths

are 750 nm (red), 600 nm (yellow), 500 nm (greenish-yellow), 430 nm (violet). Clearly such a light will appear white to the unaided eye.

Thus with white light we get a white central fringe at the point of zero path difference, followed by a few coloured fringes on its both sides, the color soon fading off to a uniform white.

In the usual interference pattern with a monochromatic source, a large number of identical interference fringes are obtained and it is usually not possible to determine the position of central maxima. Interference with white light is used to determine the position of central maxima in such cases.

The fringe visibility (v) is given by

$$v = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{max}} + I_{\text{min}}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} = \frac{2A_1 A_2}{A_1^2 + A_2^2}$$

$$\begin{array}{ll} \text{when} & I_1 = I_2 \\ \text{or} & A_1 = A_2 \\ & I_{\text{min}} = 0 \\ \text{hence,} & v = 1 \quad (\text{best visibility}) \end{array}$$

YDSE with oblique incidence

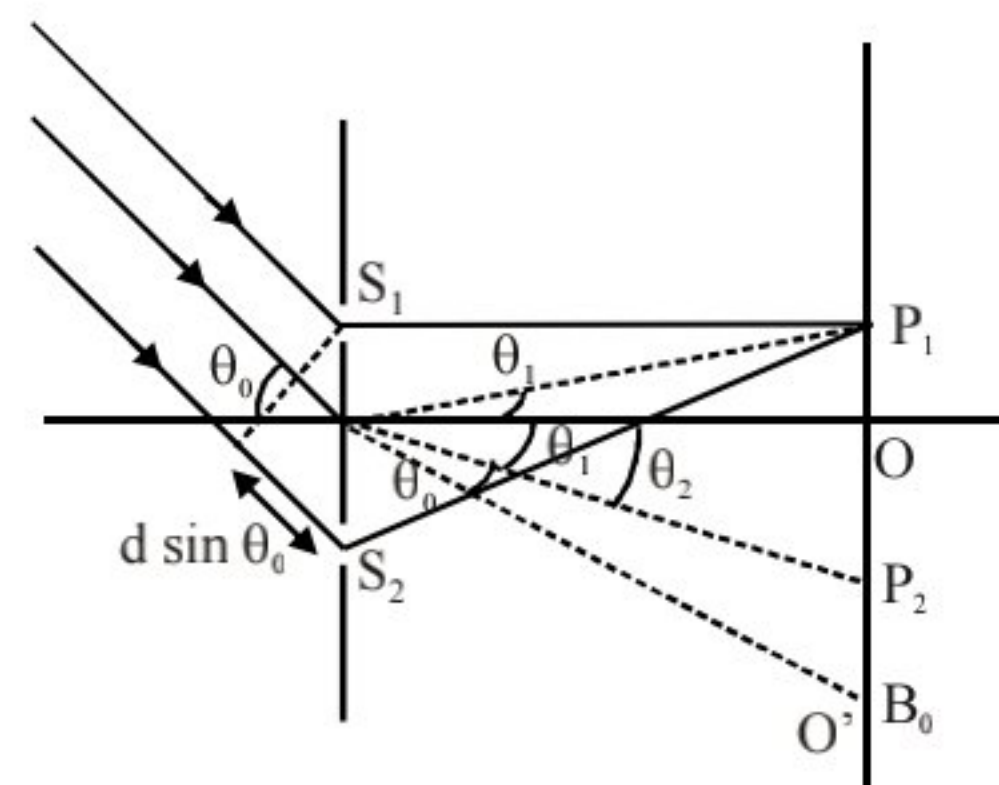
In YDSE, ray is incident on the slit at an inclination of θ_0 to the axis of symmetry of the experiment set-up for points above the central point on the screen, (say for P_1)

$$\Delta p = d \sin \theta_0 + (S_2P_1 - S_1P_1)$$

$$\Rightarrow \Delta p = d \sin \theta_0 + d \sin \theta_1 \quad (\text{if } d \ll D)$$

and for points below O on the screen, (say for P_2)

$$\begin{aligned} \Delta p &= |(d \sin \theta_0 + S_2P_2) - S_1P_2| \\ &= |(d \sin \theta_0 - (S_1P_2 - S_2P_2))| \end{aligned}$$



GEOMETRICAL PATH AND OPTICAL PATH

Actual distance travelled by light in a medium is called geometrical path (Δx). Consider a light wave given by the equation.

$$E = E_0 \sin (\omega t - kx + \phi)$$

If the light travels by Δx , its phase changes by $k\Delta x = \frac{\omega}{v} \Delta x$, where ω , the frequency of light does not depend

on the medium, but v , the speed of light depends on the medium as $v = \frac{c}{\mu}$.

Consequently, change in phase,

$$\Delta\phi = k\Delta x = \frac{\omega}{C} (\mu\Delta x)$$

It is clear that a wave travelling a distance Δx in a medium of refractive index μ suffers the same phase change as when it travels a distance $\mu\Delta x$ in vacuum. i.e. a path length of Δx in medium of refractive index μ is equivalent to a path length of $\mu\Delta x$ in vacuum.

The quantity $\mu\Delta x$ is called the optical path length of light, Δx_{opt} . And in terms of optical path length, phase difference would be given by,

$$\Delta\phi = \frac{\omega}{C} \Delta x_{\text{opt}} = \frac{2\pi}{\lambda_0} \Delta x_{\text{opt}}$$

where λ_0 = wavelength of light in vacuum.

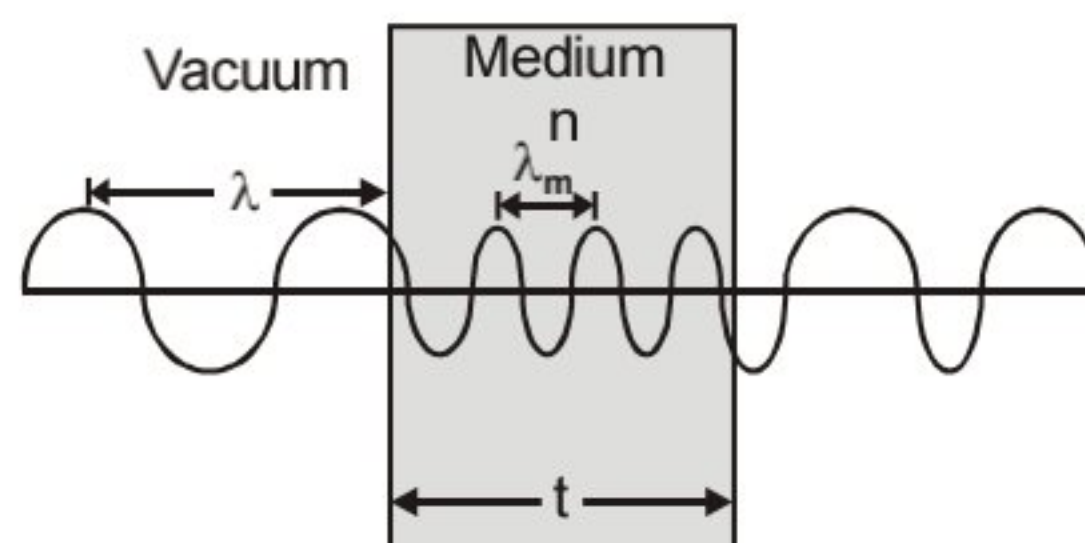
However in terms of the geometrical path length Δx ,

$$\Delta\phi = \frac{\omega}{C} (\mu\Delta x) = \frac{2\pi}{\lambda} \Delta x$$

where λ = wavelength of light in the medium $\left(\lambda = \frac{\lambda_0}{\mu} \right)$.

Equivalent optical path length

When a beam of light travels from one medium to another its speed changes but its frequency does not.



Designating the wavelength in vacuum by λ and the wavelength in the material by λ_m we have

$$\frac{\lambda}{\lambda_m} = \frac{C}{v} = n \quad \text{or} \quad \lambda_m = \frac{\lambda}{n}$$

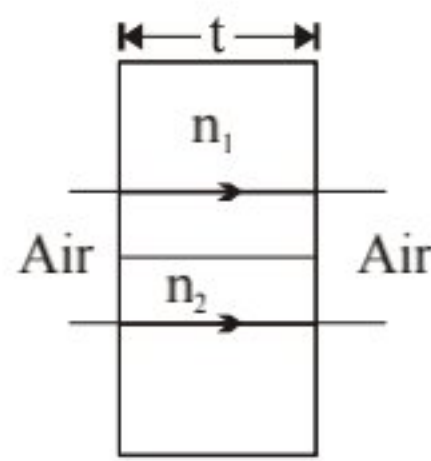
The wavelength is shorter in the medium than in vacuum. If the light beam passes through a thickness t of a medium,

$$\text{Number of wavelength in slab} = \frac{t}{\lambda_m} = \frac{t}{\lambda/n} = \frac{nt}{\lambda}$$

which shows that a thickness t of the medium has as many wavelengths as there are in a length nt of vacuum. Therefore in terms of wavelengths, a thickness t in a medium of refractive index n is equivalent to a path length nt in vacuum. The quantity nt is called equivalent optical path length.

Optical path length in terms of wave lengths

In fig. shown two light rays of identical wavelength and initially in phase in air travel through two different media of refractive indices n_1 and n_2 , same thickness t . The wavelengths of the waves will be different in the two media ; so the two waves will no longer be in phase when they emerge.



Number of wavelengths in medium 1, $N_1 = \frac{t}{\lambda_1} = \frac{t}{\lambda/n_1} = \frac{n_1 t}{\lambda}$

Number of wavelength in medium 2, $N_2 = \frac{t}{\lambda_2} = \frac{t}{\lambda/n_2} = \frac{n_2 t}{\lambda}$

To find a new phase difference we subtract the number of wavelengths of the waves in the two media ; assuming $n_2 > n_1$, we have

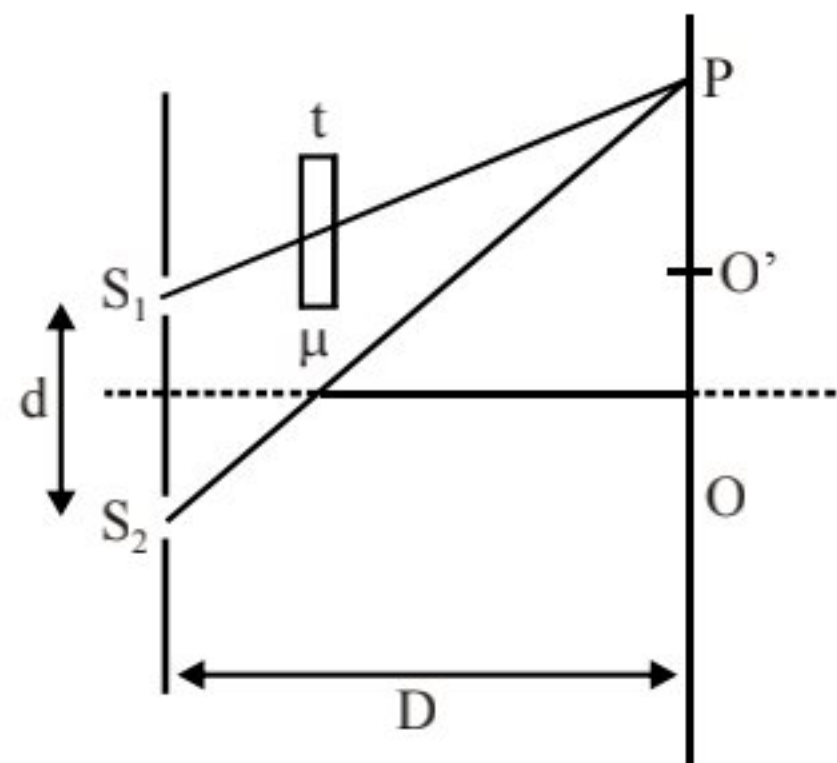
$$N_2 - N_1 = (n_2 - n_1) \frac{t}{\lambda}$$

Phase difference corresponding to difference of one wavelength is 2π ; hence phase difference corresponding to $N_2 - N_1$ is

$$\Delta\phi = \frac{2\pi}{\lambda} (n_2 - n_1) t$$

In other words we can say that the equivalent optical paths of wave in media 1 and 2 are $n_1 t$ and $n_2 t$ respectively. Thus the path difference, $\Delta x = n_2 t - n_1 t$.

1. **Displacement of fringe** on introduction of a glass slab in the path of the light coming **ou** of the slits.



On intrudction of the thin glass-slab of thickness t and refractive index μ , the optical path of the ray $S_1 P$ increases by $t(\mu - 1)$. Now the path difference between waves coming from S_1 and S_2 at any point P is

$$\begin{aligned} \Delta p &= S_2 P - (S_1 P + t(\mu - 1)) \\ &= (S_2 P - S_1 P) - t(\mu - 1) \\ \Rightarrow \Delta p &= d \sin \theta - t(\mu - 1) \text{ if } d \ll D \end{aligned}$$

$$\text{and } \Delta p = \frac{yd}{D} - t(\mu - 1) \text{ if } y \ll D \text{ as well.}$$

for central bright fringe,

$$\Delta p = 0$$

$$\Rightarrow \frac{yd}{D} = t(\mu - 1).$$

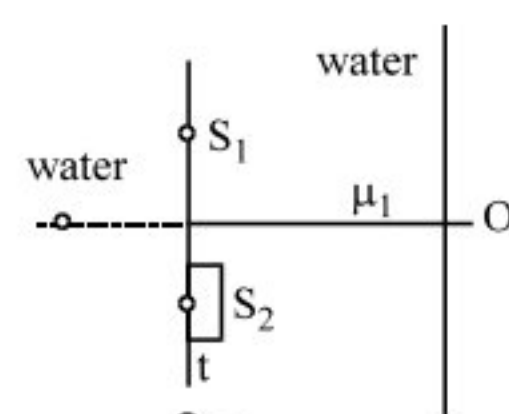
$$\Rightarrow y = OO' = (\mu - 1)t \frac{D}{d} = (\mu - 1)t \left(\frac{\beta}{\lambda} \right)$$

The whole fringe pattern gets shifted by the same distance

$$\Delta = (\mu - 1)t \frac{D}{d} = (\mu - 1)t \frac{B}{\lambda}$$

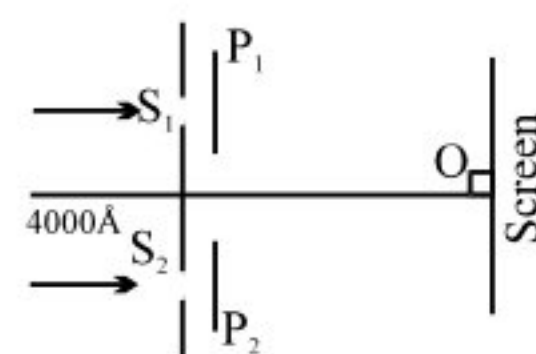
- * Notice that this shift is in the direction of the slit before which the glass slab is placed. If the glass slab is placed before the upper slit, the fringe pattern gets shifted upwards and if the glass slab is placed before the lower slit the fringe pattern gets shifted downwards.

Ex. A young's double slit experiment is conducted in water (μ_1) as shown in the figure, and a glass plate of thickness t and refractive index μ_2 is placed in the path of S_2 . Wavelength of light in water is λ . Find the magnitude of the phase difference between waves coming from S_1 and S_2 at 'O'.



(A*) $\left| \left(\frac{\mu_2}{\mu_1} - 1 \right) t \right| \frac{2\pi}{\lambda}$ (B) $\left| \left(\frac{\mu_1}{\mu_2} - 1 \right) t \right| \frac{2\pi}{\lambda}$ (C) $|(\mu_2 - \mu_1)t| \frac{2\pi}{\lambda}$ (D) $|(\mu_2 - 1)t| \frac{2\pi}{\lambda}$

Ex. P_1, P_2 are transparent plates having equal thickness $20\mu\text{m}$ and refractive indices $\mu_1 = 1.6, \mu_2 = 1.5$. P_1 transmits 75% whereas P_2 transmits 50% of energy incident. Without P_1 and P_2 intensity at O, $I_0 = 4I$. Find intensity at O after placing P_1 and P_2 . S_1, S_2 are identical slits.



Ans. $\frac{5I}{4} + I\sqrt{\frac{3}{2}}$

Sol. $S_2P - S_1P = (\mu_1 - 1)t - (\mu_2 - 1)t$
 $= 0.1t = 2 \times 10^{-6} \text{ m}$

$$\Delta\phi = \frac{\Delta x \times 2\pi}{\lambda} = \frac{2 \times 10^{-6} \times 2\pi}{4000 \times 10^{-10}} = 10\pi$$

$$I_1 = 0.75I \quad I_2 = 0.5I$$

$$I = 0.75I + 0.5I + 2(\sqrt{0.75 \times 0.5})I \cos \Delta\phi$$

$$= \frac{5I}{4} + I\sqrt{\frac{3}{2}}$$

Ex. In Young's double slit experiment the two slits are illuminated by light of wavelength $5890 \text{ \AA}$ and the distance between the fringes obtained on the screen is 0.2° . If the whole apparatus is immersed in water then the angular fringe width will be, if the refractive index of water is $4/3$

(A) 0.30° (B) 0.15° (C) 15° (D) 30°

Sol. $\omega_a = \lambda/d$

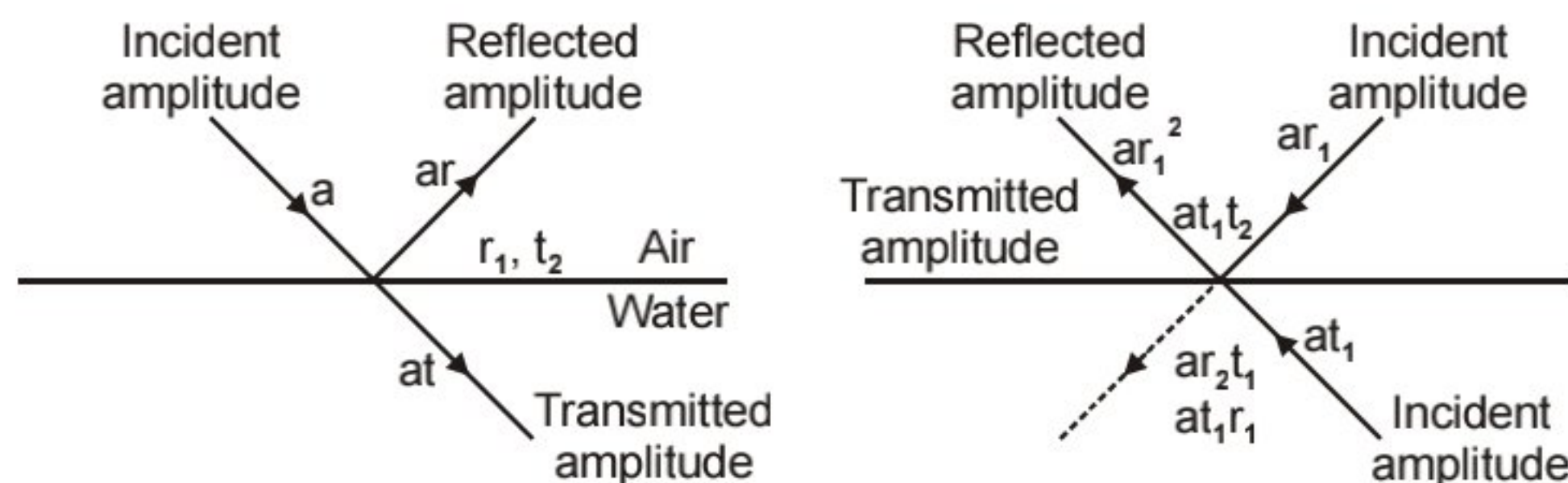
$$\therefore \omega_a \propto \lambda \Rightarrow \frac{(\omega_a)_{\text{water}}}{\omega_a} = \frac{\lambda_{\text{water}}}{\lambda}$$

$$\Rightarrow \frac{(\omega_a)_{\text{water}}}{\omega_a} = \frac{\lambda}{\mu_{\text{water}} \lambda}$$

$$\Rightarrow (\omega_a)_{\text{water}} = 0.15^\circ$$

The phase change on reflection

A ray of light is incident on air-water interface ; let the amplitude reflection and transmission coefficients be r_1 and t_1 respectively. The amplitudes of reflected and transmitted waves are ar_1 and at_1 respectively. From the principle of reversibility of light, the system retraces its whole previous motion. The wave of amplitude ar_1 gives a reflected wave of amplitude $ar_1 t_1$. The wave of amplitude at_1



gives a reflected wave of amplitude $ar_2 t_1$ and transmitted wave amplitude $at_1 t_2$.

So

$$ar_1^2 + at_1 t_2 = a$$

$$t_1 t_2 = 1 - r_1^2 \quad \dots (1)$$

Further, the waves of amplitudes $at_1 r_2$ and $ar_1 t_1$ must cancel each other.

$$at_1 r_2 + ar_1 t_1 = 0$$

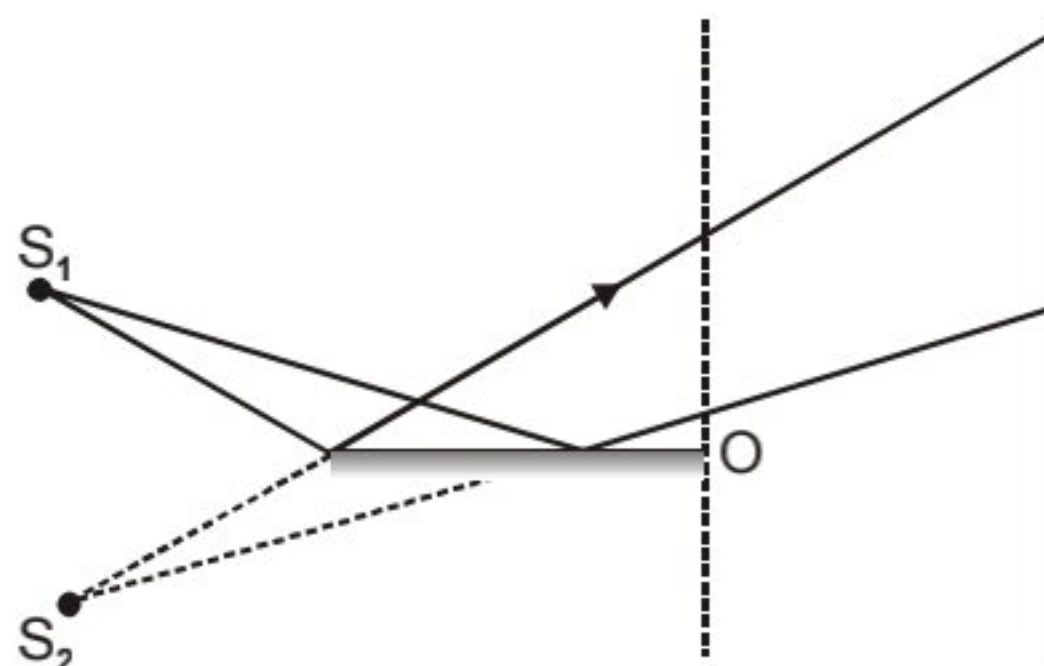
$$r_2 = -r_1 \quad \dots (2)$$

Equation shows a difference of phase of π between the two cases ; a reversal of sign means a displacement in the opposite sense. If there is no change of phase on reflection from above, there must be a phase change of π from below and vice-versa.

When light gets reflected from a denser medium there is an abrupt phase change of π ; no phase change occurs when reflection takes place from rarer medium.

The Lloyd's mirror experiment

In this arrangement the light reflected from a long mirror and the light coming directly from the source without reflection produced interference on a screen. An important feature of this experiment lies in the fact that when the screen is placed in contact with the end of the mirror the edge O of the reflecting surface comes at the centre of a dark fringe instead of a bright fringe. The direct beam does not suffer any phase change. This means that the reflected beam undergoes a phase change of π radian. Hence at a point P on the screen the conditions for minima and maxima are

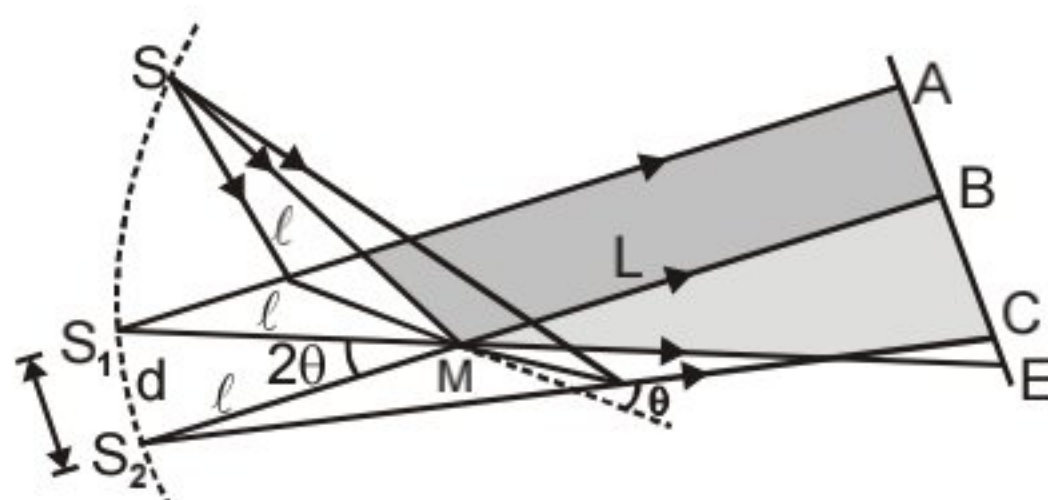


$$S_2P - S_1P = n\lambda, \quad n = 0, 1, 2, 3, \dots \text{ [minima]}$$

$$S_2P - S_1P = \left(n + \frac{1}{2}\right)\lambda \quad \text{[maxima]}$$

Fresnel's Mirrors

Figure shows Fresnel's bimirrors apparatus to produce interference by division of the wavefront. Light from a slit S is reflected by two plane mirrors slightly inclined to each other.



The mirrors produce two virtual images S_1 and S_2 of the slit, the interference fringes are observed in the region BC, where the reflected beams overlap. If θ is the angle between the planes of the mirrors, then S_1 and S_2 subtend angle 2θ at the point of intersection M between the mirrors.

If l is the distance between the slit and the mirrors intersection and L is the distance between the screen and the mirrors intersection, then the separation between the images S_1 and S_2 is

$$d = l(2\theta) = 2l\theta$$

and

$$D = l + L$$

Thus, the fringe width is given by

$$\beta = \frac{\lambda D}{d} \quad \text{or} \quad \beta = \frac{\lambda(l+L)}{2l\theta} \quad \text{or} \quad \beta = \frac{\lambda}{2\theta} \left[1 + \frac{L}{l}\right]$$

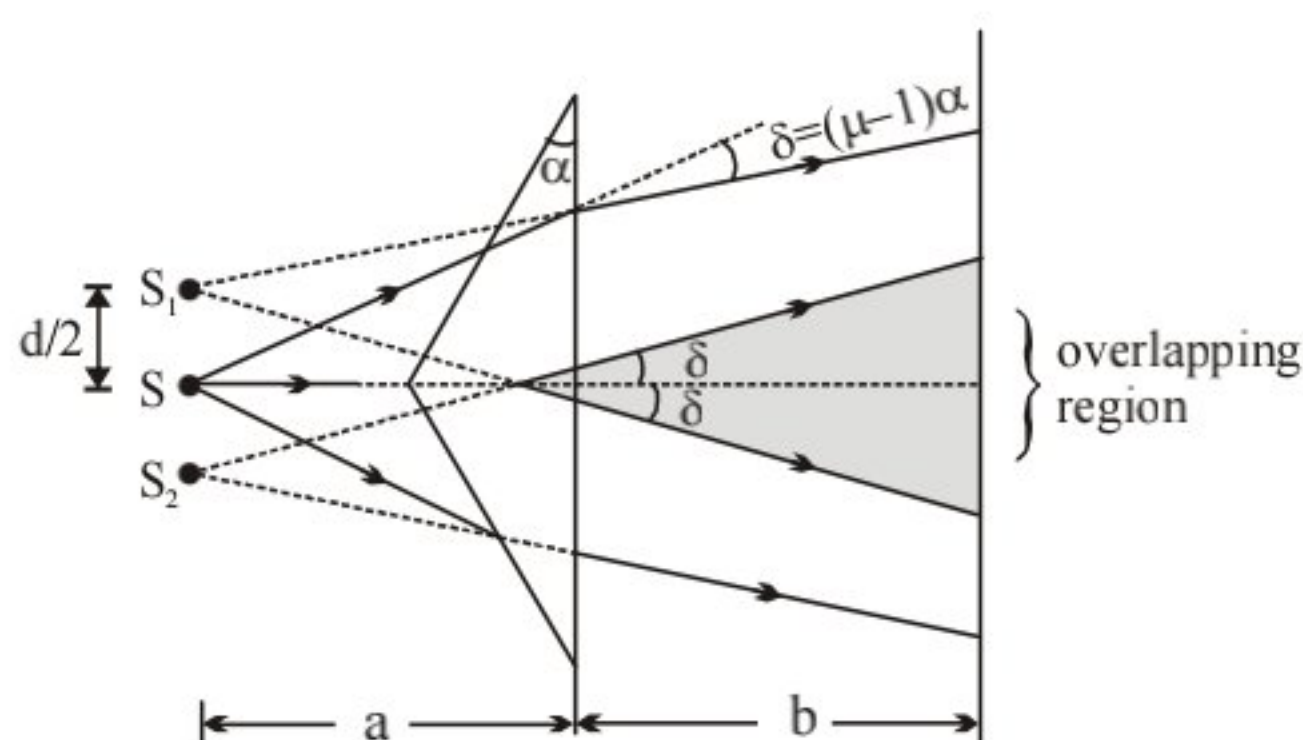
Fresnel's Biprism

Figure shows the Fresnel's biprism experiment schematically. The thin prism P refracts light from the slit source S into two beams. When a screen is placed as shown in the figure, the interference fringes are observed only in the region shown.

If A is the angle of refraction of the thin prism and μ is the refractive index of its medium, then the angle of deviation produced by the prism is

$$\delta = A(\mu - 1)$$

- (i) In numerical problems 'd' is calculated as given below :



$$\frac{d}{2} = a \tan \delta \approx a \delta = a (\mu - 1) \alpha$$

$$d = 2a (\mu - 1) \alpha$$

'd' can also be calculated using lens displacement method and it is given by

$$d = \sqrt{d_1 d_2}$$

where d_1 and d_2 are the distances between images of S_1 and S_2 in two positions of a convex lens placed between the biprism and the screen,

(ii) The expression for fringe width is same as in YDSE

$$\text{i.e., } \beta = \frac{D\lambda}{d} = \frac{(a+b)\lambda}{2a(\mu-1)\alpha}$$

and interference pattern consists of alternate bright and dark fringes.

$$\text{If source is at infinity i.e., } a \rightarrow \infty \text{ then } \beta = \frac{\lambda}{2(\mu-1)\alpha}$$

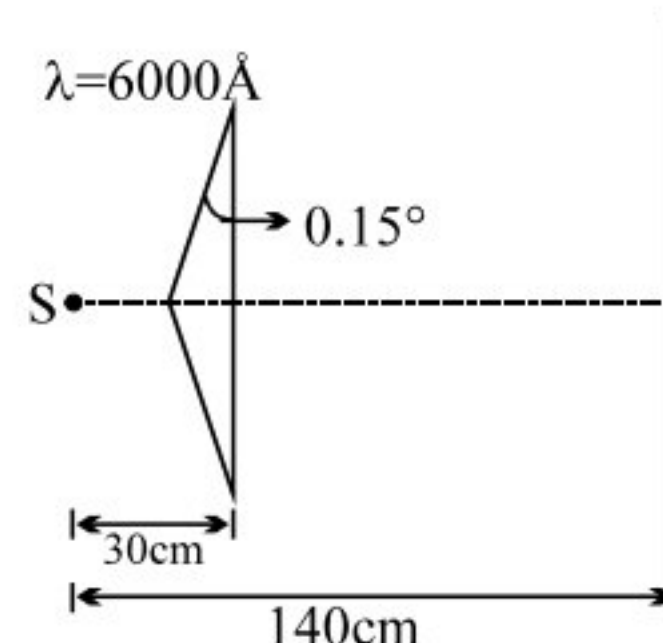
(iii) Let L = length of overlapping region from figure we have

$$\frac{L}{b} = \frac{d}{a} \quad \text{or} \quad L = \frac{bd}{a}$$

$$\text{Also, No. of fringes} = \frac{\text{Length of interference pattern}}{\text{Fringe width}} = \frac{L}{\beta}$$

Question No. 1 to 5 (5 questions)

Refracting angles of biprism are both 0.15° . Distance of biprism from source is 30 cm. Distance between source and screen is 140 cm. The Fresnel's biprism arrangement shown in the figure uses a source of light of wavelength $6000\text{\AA}$. When a converging lens is moved between the biprism and the screen two images of S are focused on the screen for two positions of the lens. The separations between the two images for the two positions are found to be 0.9 mm and 1.6 mm.



Ex. The focal length of the lens is

$$(A) \frac{120}{7} \text{ cm}$$

$$(B^*) \frac{240}{7} \text{ cm}$$

(C) Data not sufficient

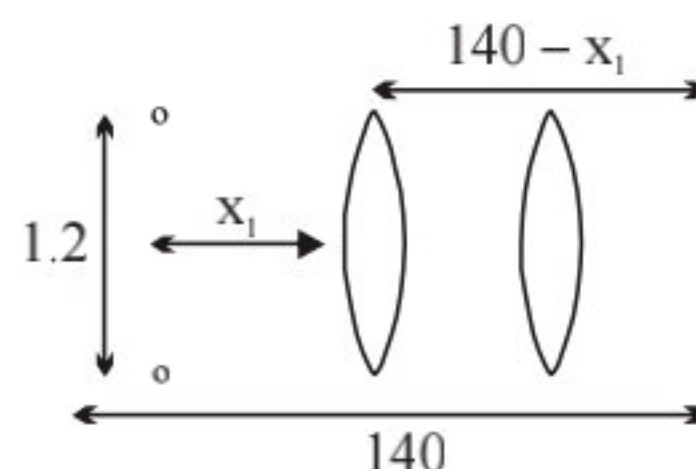
(D) None

$$\text{Sol. } h_0 = \sqrt{h_1 h_2} = \sqrt{1.6 \times 0.9} = 1.2 \text{ mm}$$

$$h_1 = 1.6$$

$$h_0 = 1.2 \quad m_1 = \frac{4}{3} = \frac{140 - x_1}{x_1} \Rightarrow x_1 = 80$$

$$\Rightarrow \frac{1}{60} - \frac{1}{-80} = \frac{1}{f} \Rightarrow f = \frac{240}{7} \text{ cm} \quad \& \quad x_2 = 60$$



Ex. Gap between two positions of the lens is

- (A) 80 cm (B) 60 cm (C) 40 cm (D*) 20 cm

Sol. $(x_2 - x_1) = 20 \text{ cm}$

Ex. Refractive index of the prism material is approximately

- (A) 1.46 (B) 1.56 (C) 1.66 (D*) 1.76

Sol. $h_0 = \sqrt{h_1 h_2} = \sqrt{1.6 \times 0.9} = 1.2 \text{ mm} = 2a\delta$

$$1.2 = 2aA(\mu - 1)$$

$$A = \frac{0.05}{180} \times \pi$$

$$a = 30 \text{ cm} \Rightarrow \mu \approx 1.76$$

Ex. The fringe width of the pattern on screen (after the lens has been removed) is

- (A*) 0.7 mm (B) 0.35 mm (C) 1.4 mm (D) None

Sol. $d = 1.2 \text{ mm}$

$$D = 140 \text{ cm}$$

$$\therefore \beta = \frac{\lambda D}{d} = 0.7 \text{ mm}$$

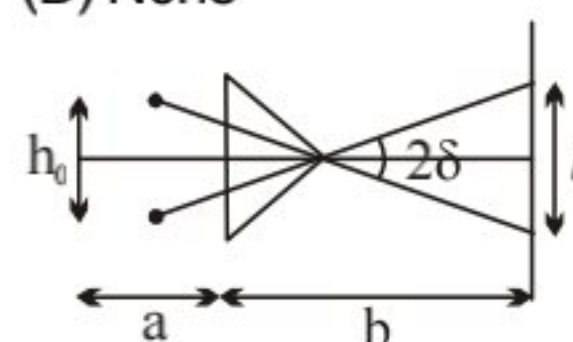
Ex. The total number of fringes obtained on the screen

- (A) $\frac{4400}{7}$ (B) $\frac{400}{7}$ (C*) $\frac{44}{7}$

Sol. $l = 2\delta b$

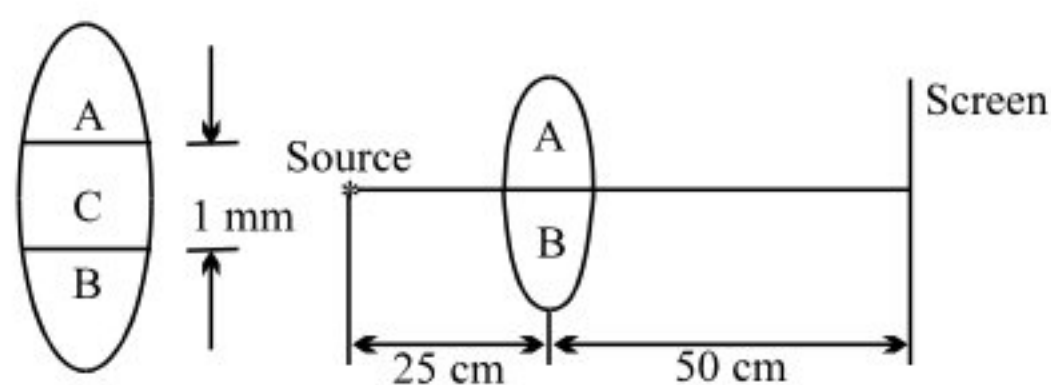
$$n = \frac{2\delta b}{\beta} = \frac{h_0 b}{a\beta} \Rightarrow \frac{44}{7}$$

(D) None



Note : Explain billet's lens arrangement.

Ex. A convex lens of focal length 50 cm is cut along the diameter into two identical halves A and B and in the process a layer C of the lens thickness 1 mm is lost. Then the two halves A and B are put together to form a composite lens. Now in front of this composite lens a source of light emitting wavelength $\lambda = 6000 \text{ \AA}$ is placed at a distance of 25 cm as shown in the figure. Behind the lens there is a screen at a distance 50 cm from it. Find the fringe width of the interference pattern obtained on the screen.



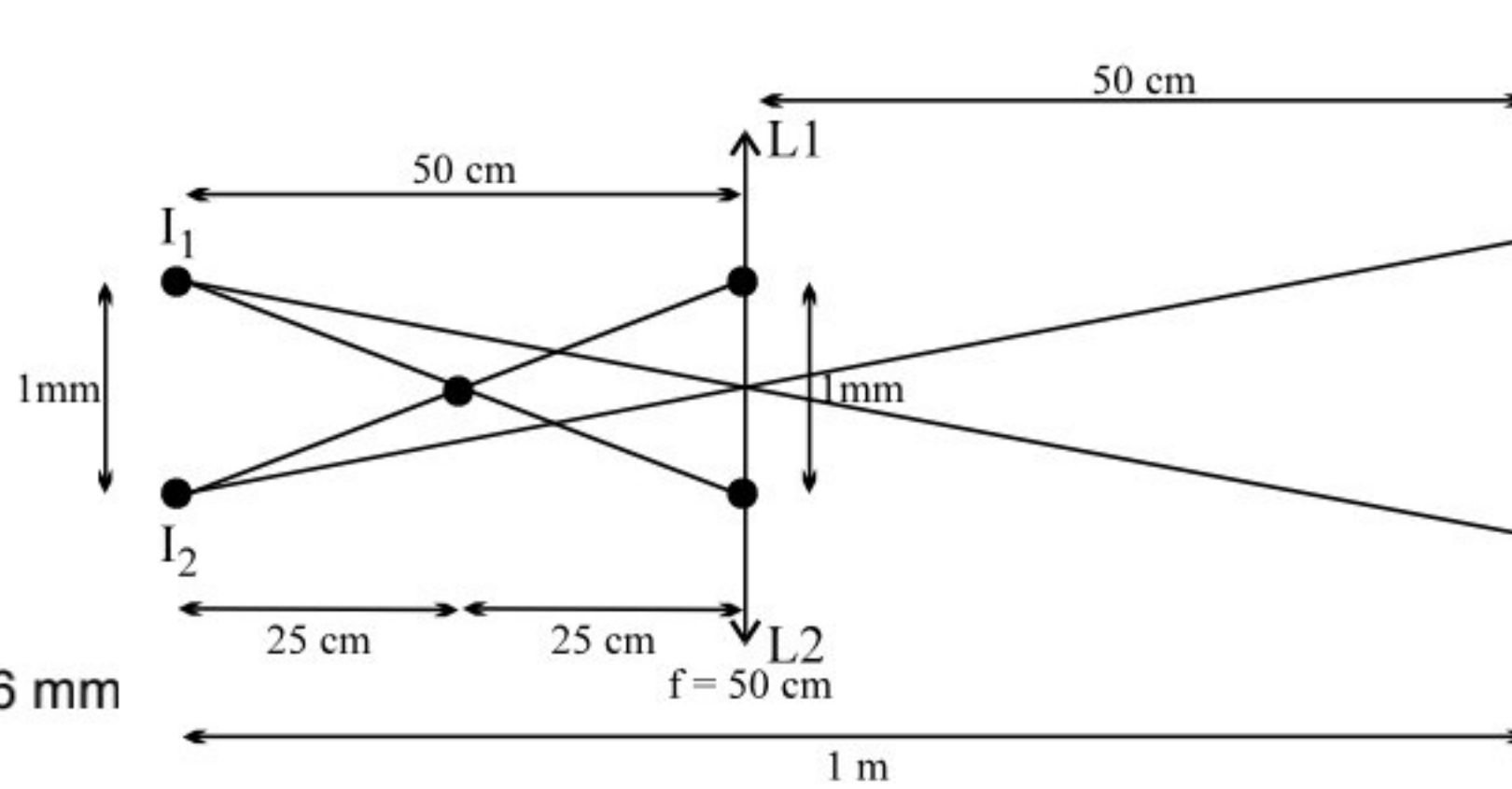
Ans. 0.6 mm

Sol. $u = -25$

$$\frac{1}{v} - \frac{1}{-25} = \frac{1}{50}$$

$$\Rightarrow \frac{1}{v} = -\frac{1}{50} \Rightarrow v = -50$$

$$\beta = \frac{6 \times 10^{-7} \times 1}{10^{-3}} = 6 \times 10^{-4} = 0.6 \text{ mm}$$

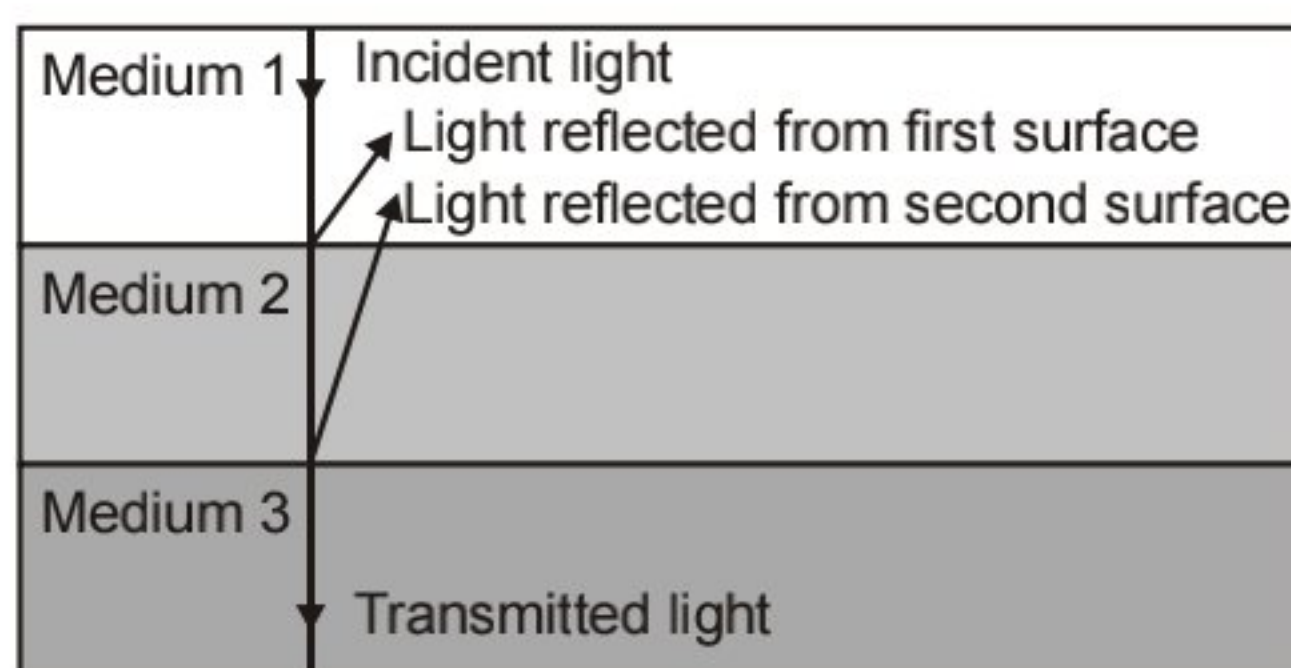


Thin film interference

When light passes the boundary between two transparent media, some light is reflected at the boundary. As shown in the figure some light is reflected from first interface and some from second interface. If we consider a monochromatic incident light the two reflected waves are also monochromatic and coherent because they arise from the same monochromatic incident light wave via amplitude division. These waves interfere, since they are superposed along the same normal line.

The phase difference between two interfering waves is due to :

- (1) Optical path difference (due to distances travelled),
- (2) Reflection from a denser medium.



The second factor is irrelevant for reflection at rarer medium.

Three situations may arise :

- (1) Neither wave experiences a phase change upon reflection.
- (2) Both the waves suffer a phase change upon reflection.

In either of these two cases the phase change due to reflection is irrelevant ; no difference in phase results due to reflection.

In either of these cases phase change is determined solely from optical path difference.

Condition for constructive interference :

$$2\mu t = m\lambda$$

Condition for destructive interference :

$$2\mu t = \left(m + \frac{1}{2}\right)\lambda$$

where $m = 0, 1, 2, \dots$

- (3) One of the reflected waves experiences a phase change of π radian upon reflection and the other waves does not.

It is material which wave suffers a phase change ; the conclusions in the previous case are first reversed.

Condition for destructive interference :

$$2\mu t = m\lambda$$

Condition for constructive interference :

$$2\mu t = \left(m + \frac{1}{2}\right)\lambda$$

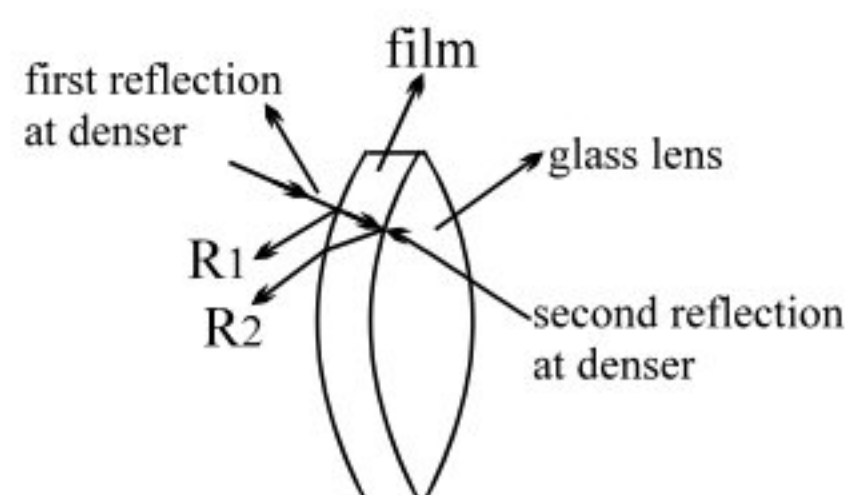
where $m = 0, 1, 2, \dots$

Ex. Many people's glasses appear to be a blue-green colour when viewed under reflected light. A thin film of index of refraction $n = 1.35$ is applied to the outside surface of the glass so that the film/glass interface does not reflect any red light incident near normal of wavelength $\lambda = 630 \text{ nm}$. What thickness must the film layer be in order to achieve this? Take the index of refractions of air and glass to be 1.0 and 1.6 respectively.

- (A) 157.5 nm (B) 315.0 nm
(C) 233.3 nm (D*) 116.7 nm

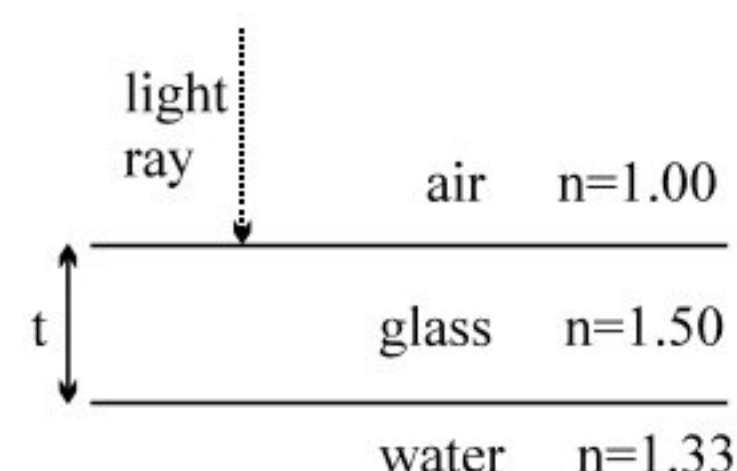
Sol. $\Delta X_{\text{net}} = 2\mu t = \lambda/2$

$$t = \frac{\lambda}{4\mu} = \frac{(630 \times 10^{-9})}{4 \times 1.35} = 116.7 \text{ nm}$$



Ex. A light ray is incident normal to a thin layer of glass. Given the figure, what is the minimum thickness of the glass that gives the reflected light an orangish color ($\lambda_{\text{air}} = 600 \text{ nm}$)?

- (A) 50 nm (B*) 100 nm (C) 150 nm
(D) 200 nm (E) 500 nm



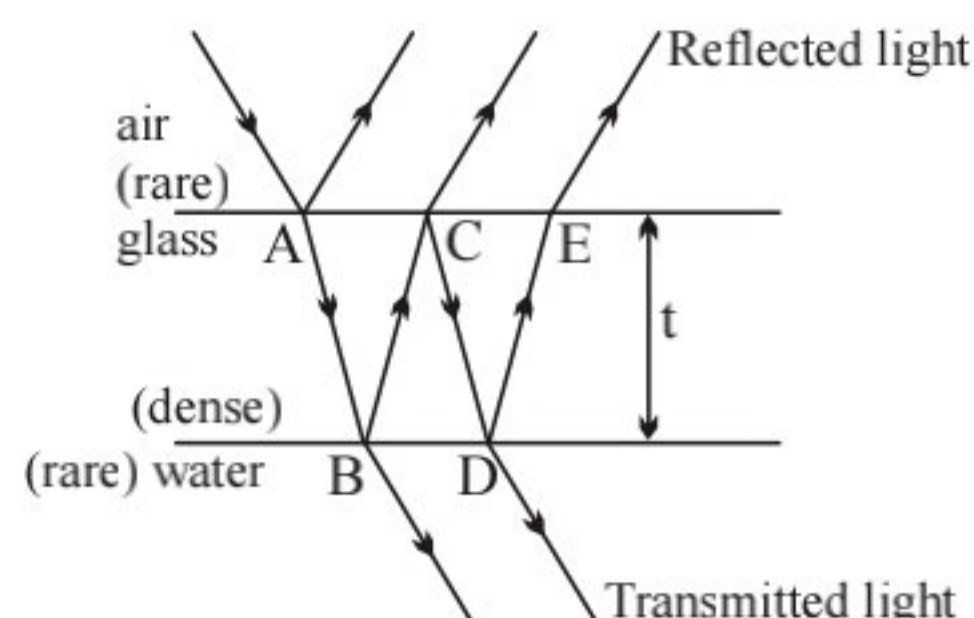
Sol. For reflected light to have orangish color, rays from A, C, E must be out of phase for $\lambda = 600 \text{ nm}$

or $\delta = (2n + 1)\pi$

or $2\mu_g t (2n + 1) \frac{\lambda}{2}$

i.e. $t = (2n + 1) \frac{\lambda}{4\mu_g}$

or $t_{\text{min}} = \frac{\lambda}{4\mu_g} = 100 \text{ nm}$



INTERFERENCE DUE TO REFLECTED LIGHT :

Consider a transparent film of thickness t and refractive index μ . A ray SA incident on the upper surface of the film is partly reflected along AR_1 and partly refracted along AB. At B part of it is reflected along BC and finally emerges out along CR_2 . The difference in path between the two rays. AR_1 and CR_2 is calculated as given below :

Let CN and BM be perpendicular to AR_1 and AC. As the paths of the rays AR_1 and CR_2 beyond CN are equal. The path difference between them is

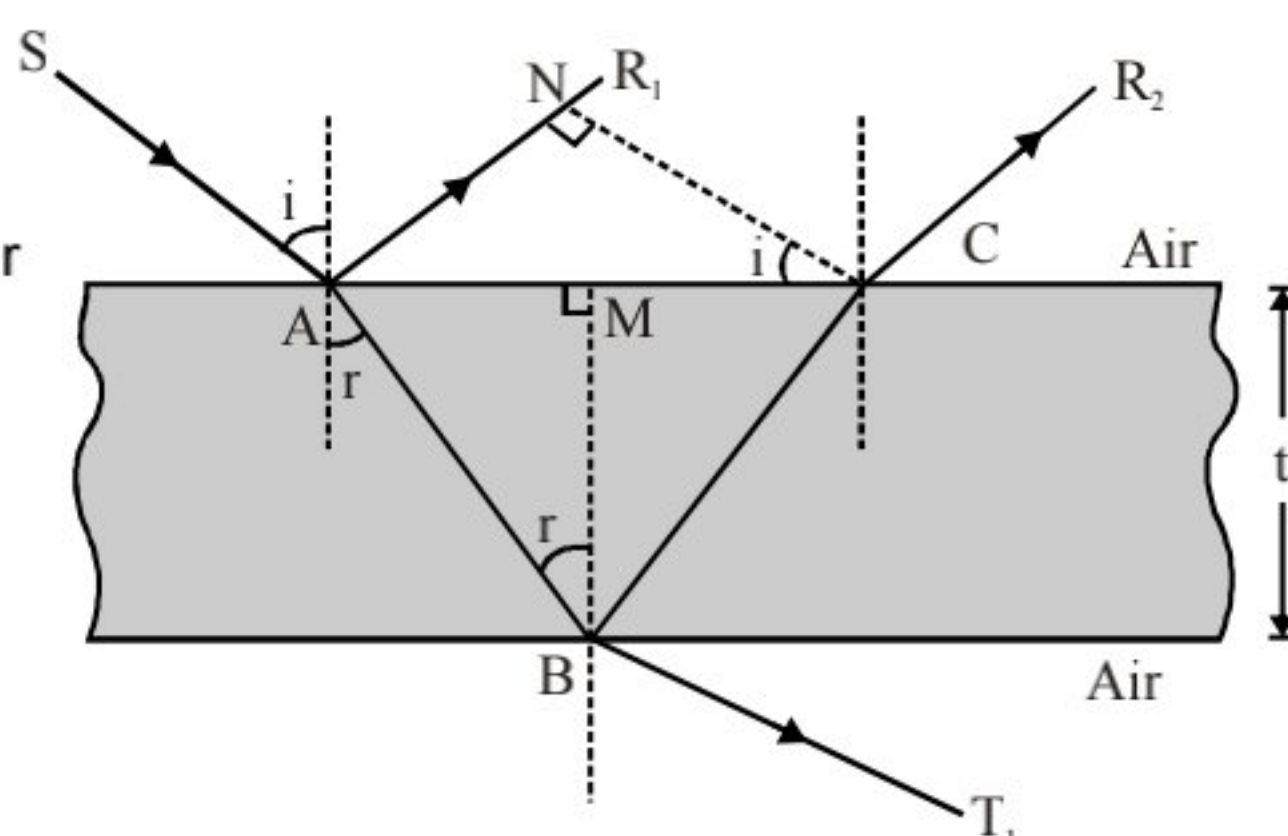
$$\Delta x = \text{Path ABC in film} - \text{Path AN in air} \\ = \mu(AB + BC) - AN = 2\mu AB - AN$$

Now, $AB = BC = \frac{AB}{BM} \times BM = BM \sec r = t \sec r$

and $AN = \frac{AN}{AC} \cdot AC = AC \sin i = 2 AM \sin i$

$$= 2 \frac{AM}{BM} BM \sin i = 2 (\tan r) t \left(\frac{\sin i}{\sin r} \sin r \right)$$

$$= 2\mu t \frac{\sin^2 r}{\cos r} = 2\mu t \sec r \sin^2 r$$



Then, $\Delta = 2\mu AB - AN = 2\mu t \sec r - 2\mu t \sec r \sin^2 r = 2\mu t \sec r (1 - \sin^2 r) = 2\mu t \cos r$

The ray AR_1 having suffered a reflection at the surface of denser medium undergoes a phase change π or

path diff. of $\frac{\lambda}{2}$.

At B the reflection takes place when the ray is going from a denser to rarer medium and there is no phase change.

Hence, the effective path difference between AR_1 and CR_2 is given by

$$\text{Path Diff. } (\Delta x) = 2\mu t \cos r - \frac{\lambda}{2}$$

- (i) If the path difference $\Delta x = n\lambda$ where $n = 0, 1, 2, 3, 4$ etc., constructive interference takes place and the film appears bright.

$$\therefore 2\mu t \cos r - \frac{\lambda}{2} = n\lambda \quad \text{or} \quad 2\mu t \cos r = (2n + 1) \frac{\lambda}{2}$$

- (ii) If the path difference $\Delta x = (2n + 1) \frac{\lambda}{2}$ where $n = 0, 1, 2, \dots$ etc., destructive interference takes place and the film appears dark.

$$\therefore 2\mu t \cos r - \frac{\lambda}{2} = (2n + 1) \frac{\lambda}{2} \quad \text{or} \quad 2\mu t \cos r = n\lambda$$

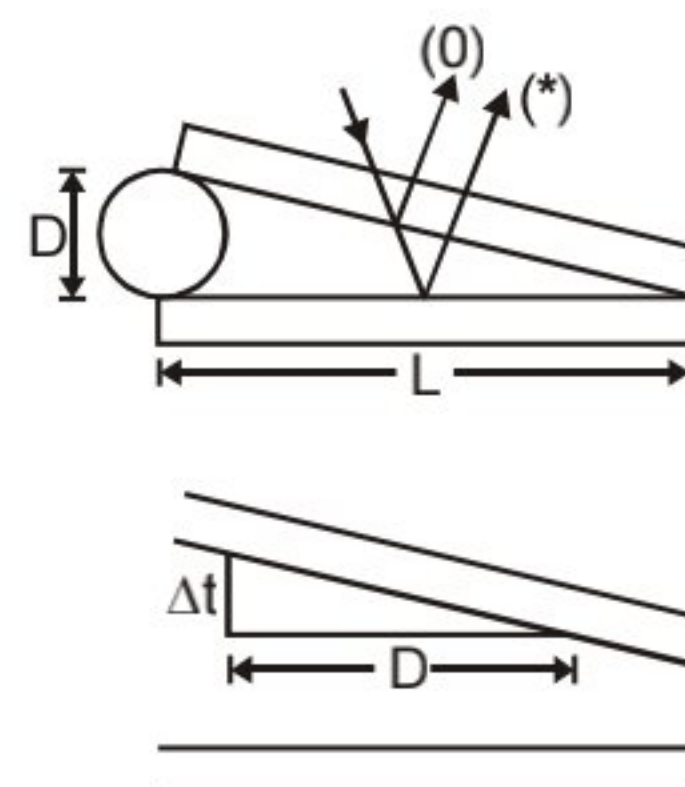
Remarks :

- (i) If the thickness of the film is very small as compared to the wavelength of light used, so that $2\mu t \cos r$ can be neglected, then the total path difference between AR_1 and CR_2 will reduce to $\frac{\lambda}{2}$. Thus two rays will interfere destructively and darkness will result.
- (ii) It should be remembered that the interference pattern will not be perfect because the intensities of the ray AR_1 and CR_2 will not be the same and their amplitudes are

Fringes of equal thickness

Soap bubbles and oil films on a road do not have uniform thickness of the film at any given point determines whether the reflected light has a maximum or minimum intensity. When white light is used, each wavelength has its own fringe pattern. At a given point of the film, one wavelength may be enhanced and / or another wavelength suppressed. This is the source of the colors in soap bubbles and oil films on the road.

A wedge-shaped film of air may be produced by placing a sheet of paper or a hair between the ends of two glass plates, as in fig. With flat plates, one sees a series of bright and dark bands, each characteristic of a particular thickness. If the plates are not flat, the fringes are not straight; each is locus of points with the same thickness. If one plate is known to be flat, the fringes display the irregularities of the other, as shown in figure. The pattern shows where the plate needs to be polished for it to be made "optically flat."



Ex. A wedge-shaped film of air is produced by placing a fine wire of diameter D between the ends of two flat glass plates of length $L = 20$ cm, as in fig. When the air film is illuminated with light of wavelength $\lambda = 550$ nm, there are 12 dark fringes per centimeter. Find D .

Sol. As indicated in fig. only one of the reflected ray suffers a phase inversion. At the thin end of the wedge, where the thickness is less than $\lambda/4$, the two rays interfere destructively. This region is dark in the reflected light. The condition for destructive interference in the reflected light is

$$2t = m\lambda \quad m = 0, 1, 2, \dots$$

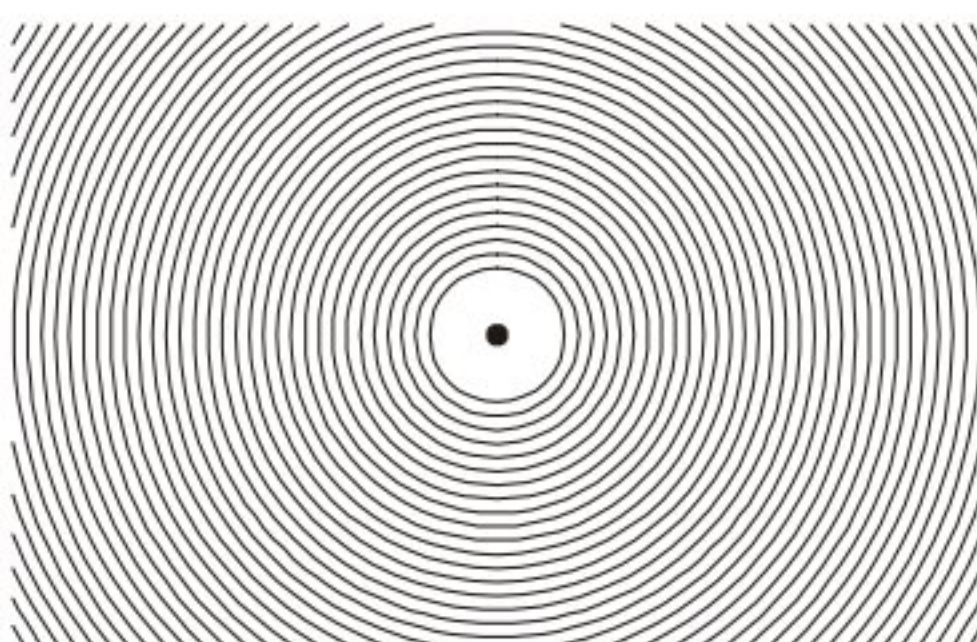
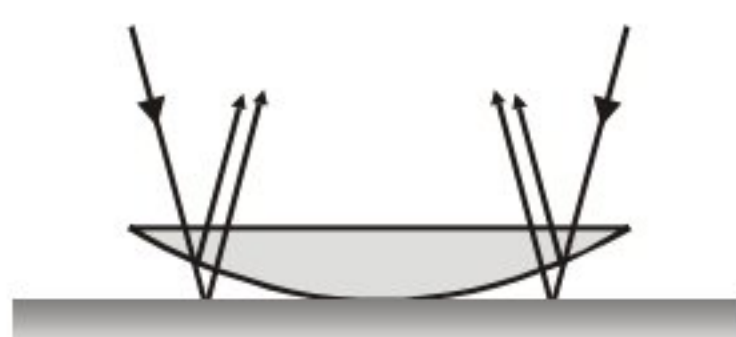
the change in thickness between adjacent dark fringes is $\Delta t = \lambda/2$. The horizontal spacing between fringes $d = 1/12$ cm $= 8.3 \times 10^{-4}$ m. From figure we see that $D/L = \Delta t/\delta$, so

$$\Delta = \frac{\lambda L}{2d} = \frac{(5.5 \times 10^{-7} \text{ m})(0.2 \text{ m})}{16.6 \times 10^{-4} \text{ m}}$$

Thus $D = 6.6 \times 10^{-5}$ m

Newton's Rings

When a lens with a large radius of curvature is placed on a flat plate, as to fig. a thin film of air is formed. When the film is illuminated with monochromatic light, circular fringes, called newton's rings, can be with the unaided eye or with a low power microscope (figure). An important feature of Newton's rings is the dark central spot. Newton tried polishing the surfaces to get rid of it, The dark spot was also initially puzzling to Young. It implied that the light wave suffers a phase inversion on reflection at a medium with a higher refractive index. Young tested this idea by placing oil of sassafras between a lens of crown glass and a plate of flint glass. The refractive index of the oil is between the values for these two glasses. Since both reflections occur at a medium with a higher refractive index, they should both suffer a phase inversion and therefore be in phase. This is precisely what happened : The central spot became bright and undoubtedly gave Young much satisfaction.



Ex. In an experiment on Newton's rings the light has a wavelength of 600 nm. The lens has a refractive index of 1.5 and a radius of curvature of 2.5 m. Find the radius of the 5th bright fringe.

Sol. If R is the radius of curvature of the lens, then from fig. we see that $r^2 = R^2 - (R - t)^2$, where r is the radius of a fringe and t is the thickness of the film. Since t is very small, we may drop terms in t^2 to obtain.

$$r^2 \approx 2Rt \quad \dots (i)$$



In order to find r , we must first find t . The condition for a bright fringe is

$$2t = \left(m + \frac{1}{2}\right)\lambda_F \quad \dots (ii)$$

We note that $n = 1$ for the air film (the index for the glass is irrelevant) and that $m = 4$ for the fifth bright fringe. Thus, from (ii)

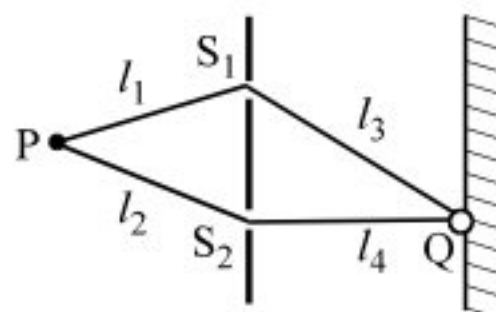
$$t = \frac{(4.5)(6 \times 10^{-7})}{2} = 1.35 \times 10^{-6} \text{ m}$$

Substituting this into (i), we find $r = \sqrt{2Rt} = 2.6 \times 10^{-3} \text{ m}$.



EXERCISE-I

- Two coherent monochromatic light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are :
 (A) $5I$ and I (B) $5I$ and $3I$ (C) $9I$ and I (D) $9I$ and $3I$
- Figure shown plane waves refracted for air to water using Huygen's principle
 a, b, c, d, e are lengths on the diagram. The refractive index of water wrt air is the ratio.
 (A) a/e (B) b/e (C) b/d (D) d/b
- When light is refracted into a denser medium,
 (A) its wavelength and frequency both increases
 (B) its wavelength increase but frequency remains unchanged
 (C) its wavelength decrease but frequency remains unchanged
 (D) its wavelength and frequency both decrease.
- In a double slit experiment, instead of taking slits of equal widths, one slit is made twice as wide as the other. Then, in the interference pattern
 (A) the intensities of both the maxima and the minima increase.
 (B) the intensity of the maxima increases and the minima has zero intensity.
 (C) the intensity of the maxima decreases and that of the minima increases.
 (D) the intensity of the maxima decreases and the minima has zero intensity.
- Two monochromatic (wavelength = $a/5$) and coherent sources of electromagnetic waves are placed on the x-axis at the points $(2a, 0)$ and $(-a, 0)$. A detector moves in a circle of radius $R (> 2a)$ whose centre is at the origin. The number of maxima detected during one circular revolution by the detector are
 (A) 60 (B) 15 (C) 64 (D) None
- In YDSE how many maxima can be obtained on the screen if wavelength of light used is 200nm & $d = 700\text{ nm}$
 (A) 12 (B) 7 (C) 18 (D) none of these
- In a YDSE, the central bright fringe can be identified :
 (A) as it has greater intensity than the other bright fringes.
 (B) as it is wider than the other bright fringes.
 (C) as it is narrower than the other bright fringes.
 (D) by using white light instead of single wavelength light.
- If the Young's double slit experiment is performed with white light, then which of the following is not true.
 (A) the central maximum will be white (B) there will not be a completely dark fringe
 (C) the bright fringe next to the central will be red (D) the bright fringe next to the central will be violet
- Two identical narrow slits S_1 and S_2 are illuminated by light of wavelength λ from a point source P.

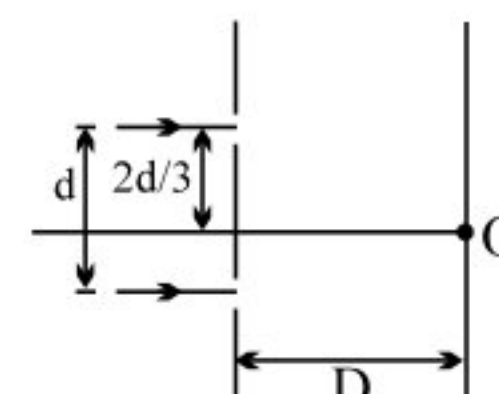
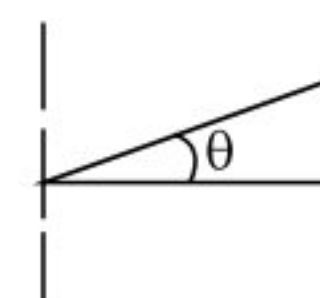
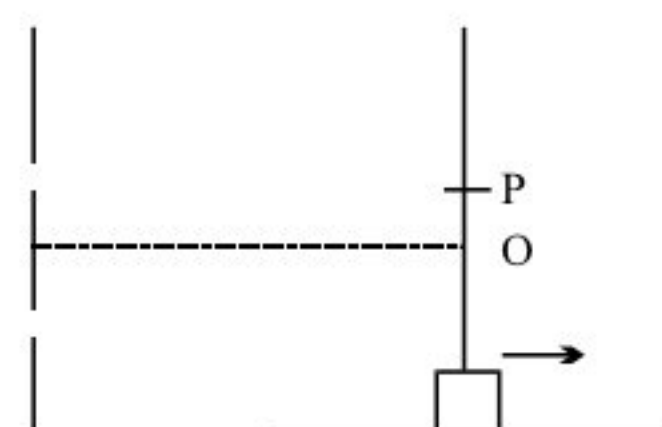


If, as shown in the diagram above the light is then allowed to fall on a screen, and if n is a positive integer, the condition for destructive interference at Q is that

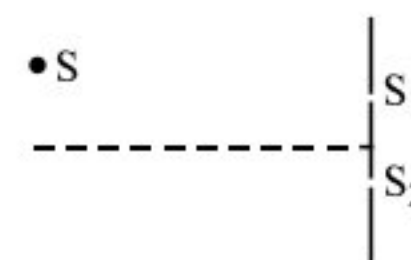
- | | |
|--|---|
| (A) $(l_1 - l_2) = (2n + 1)\lambda/2$ | (B) $(l_3 - l_4) = (2n + 1)\lambda/2$ |
| (C) $(l_1 + l_2) - (l_3 + l_4) = n\lambda$ | (D) $(l_1 + l_3) - (l_2 + l_4) = (2n + 1)\lambda/2$ |



10. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ . In another experiment with the same setup the two slits are sources of equal amplitude A and wavelength λ but are incoherent. The ratio of the average intensity of light at the midpoint of the screen in the first case to that in the second case is
 (A) 1 : 1 (B) 2 : 1 (C) 4 : 1 (D) none of these
11. In a Young's double slit experiment, a small detector measures an intensity of illumination of I units at the centre of the fringe pattern. If one of the two (identical) slits is now covered, the measured intensity will be
 (A) $2I$ (B) I (C) $I/4$ (D) $I/2$
12. In a young double slit experiment D equals the distance of screen and d is the separation between the slit. The distance of the nearest point to the central maximum where the intensity is same as that due to a single slit, is equal to
 (A) $\frac{D\lambda}{d}$ (B) $\frac{D\lambda}{2d}$ (C) $\frac{D\lambda}{3d}$ (D) $\frac{2D\lambda}{d}$
13. In a Young's Double slit experiment, first maxima is observed at a fixed point P on the screen. Now the screen is continuously moved away from the plane of slits. The ratio of intensity at point P to the intensity at point O (centre of the screen)
 (A) remains constant
 (B) keeps on decreasing
 (C) first decreases and then increases
 (D) First decreases and then becomes constant
14. Two slits are separated by 0.3 mm . A beam of 500 nm light strikes the slits producing an interference pattern. The number of maxima observed in the angular range $-30^\circ < \theta < 30^\circ$.
 (A) 300 (B) 150
 (C) 599 (D) 149
15. In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the white spot on the screen from O is [Assume $d \ll D$, $\lambda \ll d$]
 (A) 0 (B) $d/2$
 (C) $d/3$ (D) $d/6$
16. In the above question if the light incident is monochromatic and point O is a maxima, then the wavelength of the light incident cannot be
 (A) $d^2/3D$ (B) $d^2/6D$ (C) $d^2/12D$ (D) $d^2/18D$
17. In Young's double slit arrangement, water is filled in the space between screen and slits. Then :
 (A) fringe pattern shifts upwards but fringe width remains unchanged.
 (B) fringe width decreases and central bright fringe shifts upwards.
 (C) fringe width increases and central bright fringe does not shift.
 (D) fringe width decreases and central bright fringe does not shift.
18. Light of wavelength λ in air enters a medium of refractive index μ . Two points in this medium, lying along the path of this light, are at a distance x apart. The phase difference between these points is :
 (A) $\frac{2\pi\mu x}{\lambda}$ (B) $\frac{2\pi x}{\mu\lambda}$ (C) $\frac{2\pi(\mu-1)x}{\lambda}$ (D) $\frac{2\pi x}{(\mu-1)\lambda}$

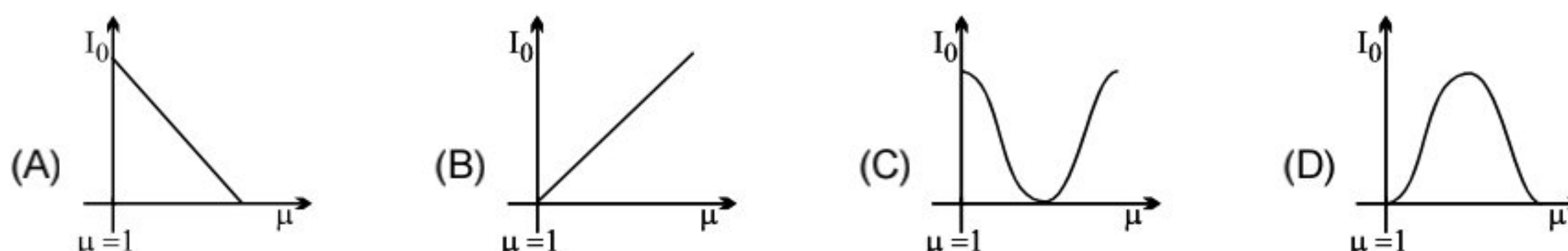


19. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits so that it is closer to the upper slit, as shown. Then,



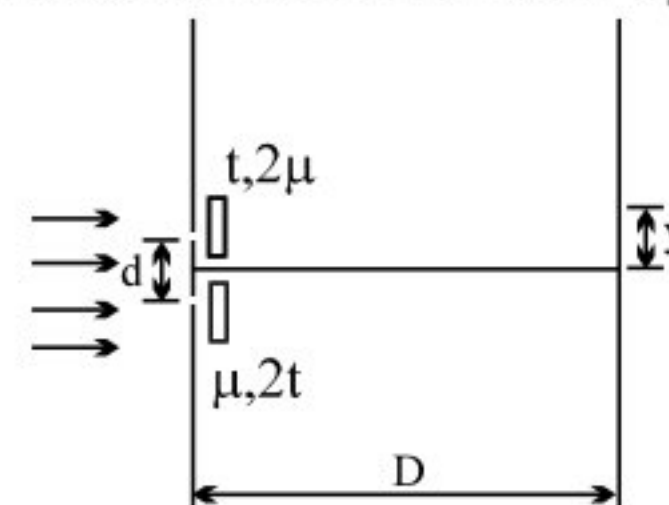
- (A) the fringe width will increase and fringe pattern will shift down.
 (B) the fringe width will remain same but fringe pattern will shift up.
 (C) the fringe width will decrease and fringe pattern will shift down.
 (D) the fringe width will remain same but fringe pattern will shift down.

20. In a YDSE experiment if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with ' μ ' will be best represented by ($\mu \geq 1$). [Assume slits of equal width and there is no absorption by slab]



21. In the YDSE shown the two slits are covered with thin sheets having thickness t & $2t$ and refractive index 2μ and μ . Find the position (y) of central maxima

- (A) zero
 (B) $\frac{tD}{d}$
 (C) $-\frac{tD}{d}$
 (D) None



22. Two monochromatic and coherent point sources of light are placed at a certain distance from each other in the horizontal plane. The locus of all those points in the horizontal plane which have construct interference will be

- (A) a hyperbola
 (B) family of hyperbolas
 (C) family of straight lines
 (D) family of parabolas

23. A thin film of thickness t and index of refraction 1.33 coats a glass with index of refraction 1.50. What is the least thickness t that will strongly reflect light with wavelength 600 nm incident normally?

- (A) 225 nm
 (B) 300 nm
 (C) 400 nm
 (D) 450 nm

24. It is necessary to coat a glass lens with a non-reflecting layer. If the wavelength of the light in the coating is λ , the best choice is a layer of material having an index of refraction between those of glass and air and a thickness of

- (A) $\frac{\lambda}{4}$
 (B) $\frac{\lambda}{2}$
 (C) $\frac{3\lambda}{8}$
 (D) λ

25. A parallel coherent beam of light falls on fresnel biprism of refractive index μ and angle α . The fringe width on a screen at a distance D from biprism will be (wavelength = λ)

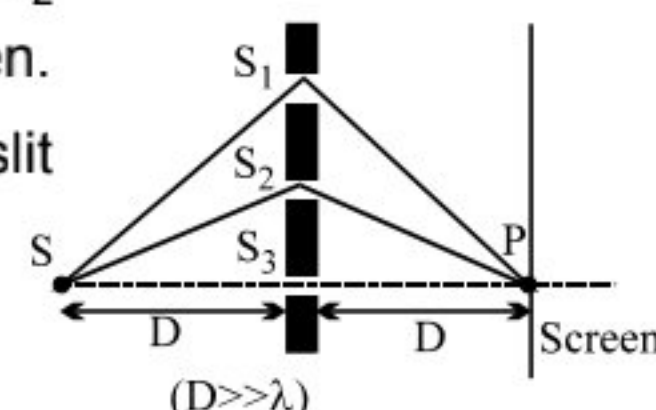
- (A) $\frac{\lambda}{2(\mu-1)\alpha}$
 (B) $\frac{\lambda D}{2(\mu-1)\alpha}$
 (C) $\frac{D}{2(\mu-1)\alpha}$
 (D) none



26. In Young's double slit experiment, the wavelength of red light is $7800 \text{ \AA}$ and that of blue light is $5200 \text{ \AA}$. The value of n for which n^{th} bright band due to red light coincides with $(n + 1)^{\text{th}}$ bright band due to blue light, is :
 (A) 1 (B) 2 (C) 3 (D) 4

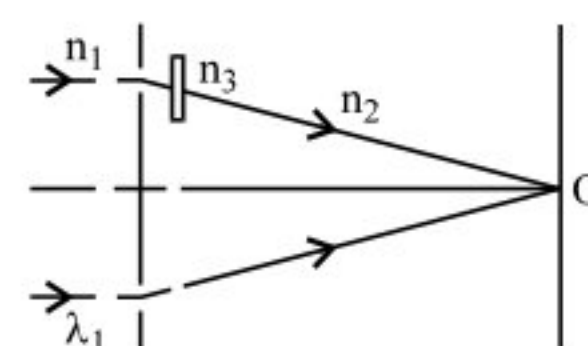
27. The ratio of the intensity at the centre of a bright fringe to the intensity at a point one-quarter of the fringe width from the centre is
 (A) 2 (B) $1/2$ (C) 4 (D) 16

28. A monochromatic light source of wavelength λ is placed at S. Three slits S_1 , S_2 and S_3 are equidistant from the source S and the point P on the screen. $S_1P - S_2P = \lambda/6$ and $S_1P - S_3P = 2\lambda/3$. If I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is
 (A) $3I$ (B) $5I$
 (C) $8I$ (D) zero



29. In the figure shown in YDSE, a parallel beam of light is incident on the slit from a medium of refractive index n_1 . The wavelength of light in this medium is λ_1 . A transparent slab of thickness 't' and refractive index n_3 is put in front of one slit. The medium between the screen and the plane of the slits is n_2 . The phase difference between the light waves reaching point 'O' (symmetrical, relative to the slits) is :

- (A) $\frac{2\pi}{n_1\lambda_1} (n_3 - n_2) t$ (B) $\frac{2\pi}{\lambda_1} (n_3 - n_2) t$
 (C) $\frac{2\pi n_1}{n_2 \lambda_1} \left(\frac{n_3}{n_2} - 1 \right) t$ (D) $\frac{2\pi n_1}{\lambda_1} (n_3 - n_1) t$



30. In a YDSE with two identical slits, when the upper slits is covered with a thin, perfectly transparent sheet of mica, the intensity at the centre of screen reduces to 75% of the initial value. Second minima is observed to be above this point and third maxima below it. Which of the following can not be a possible value of phase difference caused by the mica sheet

- (A) $\frac{\pi}{3}$ (B) $\frac{13\pi}{3}$
 (C) $\frac{17\pi}{3}$ (D) $\frac{11\pi}{3}$

31. Match each of the following descriptions to one term from the list at left :

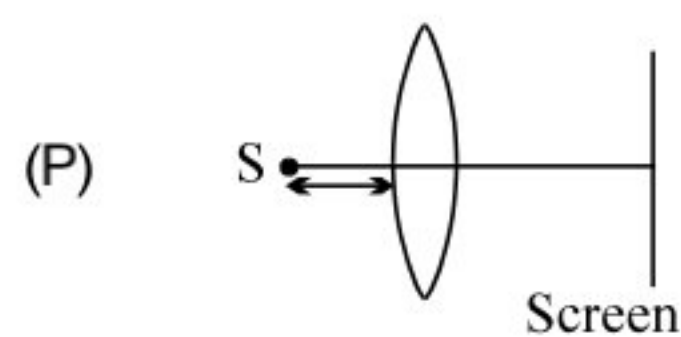
Column-I	Column-II
(A) Dispersion	(P) differing index of refraction for different wavelengths of light
(B) Huygens's principle	(Q) addition of waves with slightly different frequency at the same point in space
(C) Beats	(R) decrease in the speed of light in materials with $n > 1$
(D) Refraction	(S) any point on a propagating wavefront acts as the source of new "wavelets"

32. Light from source S ($|u| < |f|$) falls on lens and screen is placed on the other side. The lens is formed by cutting it along principal axis into two equal parts and are joined as indicated in column II.

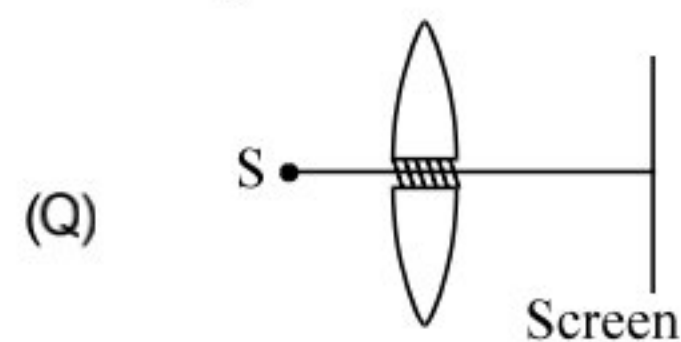
Column I

- (A) Plane of image move towards screen if $|f|$ is increased
- (B) Images formed will be virtual
- (C) Separation between images increase if $|u|$ decreases
- (D) Interference pattern can be obtained if screen is suitably positioned.

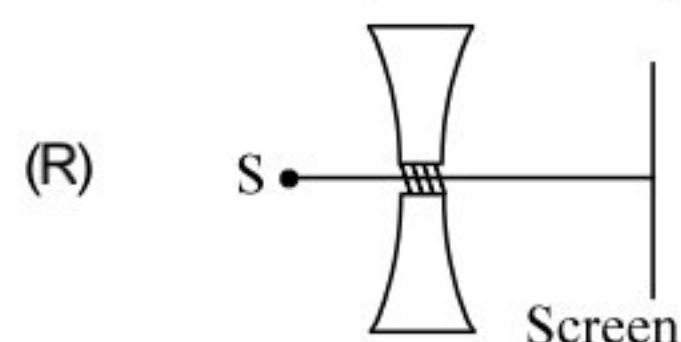
Column II



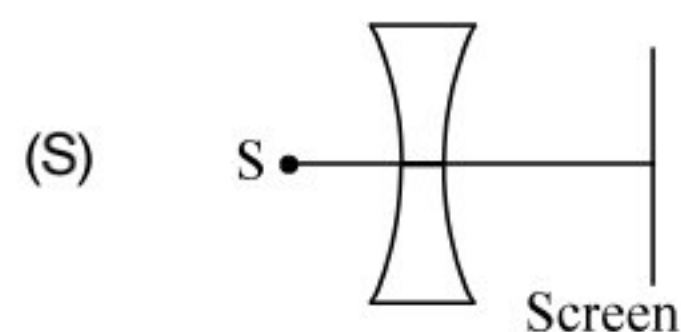
Small portion of each part near pole is removed. The remaining parts are joined.



The two parts are separated slightly. The gap is filled by opaque material.



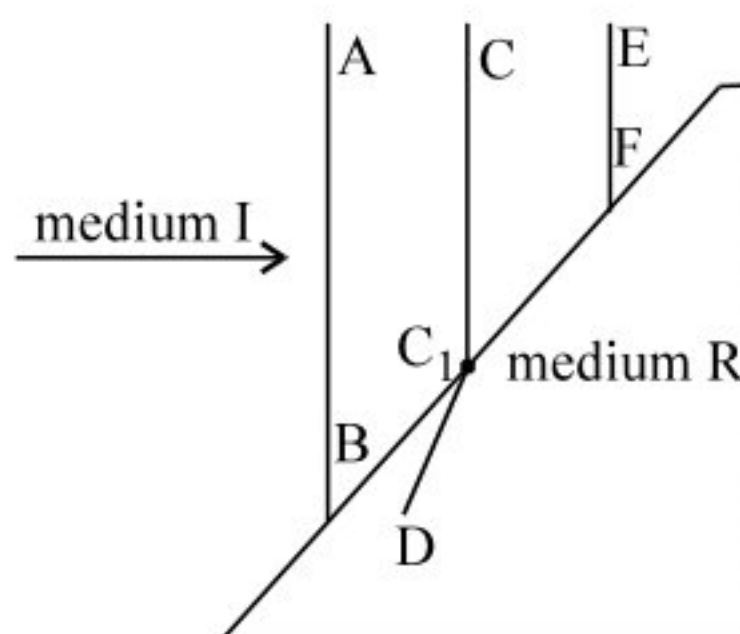
The two parts are separated slightly. The gap is filled by opaque material.



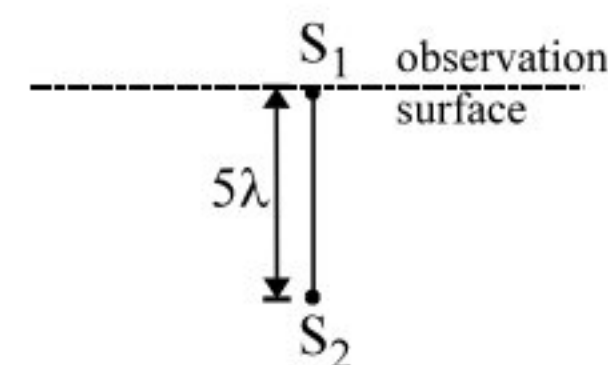
Small portion of each part near pole is removed. The remaining parts are joined.

EXERCISE-II

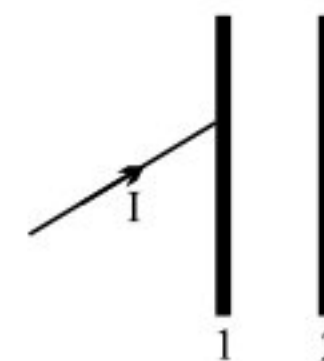
1. The diagram shows three wavefronts incident light on a boundary between medium I and medium R. Wavefront CD is shown crossing the boundary. Wavefront EF is incomplete.



- (i) On the diagram above, draw a line to complete the wavefront EF. Explain in one sentence how the line is to be drawn?
 - (ii) Explain in which medium, I or R, the wave has the higher speed.
2. In Young's double-slit experiment the two slits act as coherent sources of equal amplitude 'A' and of wavelength ' λ '. In another experiment with the same set up the two slits are sources of equal amplitude 'A' and wavelength ' λ ', but are incoherent. The ratio of the intensity of light at the midpoint of the screen in the first case to that in the second case is _____.
3. The two slits in YDSE are separated by a distance of 0.03 cm. An interference pattern is produced at a screen 1.5 m away. The fourth bright fringe is at a distance of 1 cm from the central maxima. Determine the wavelength of the light used.
4. In Young's double slit experiment the slits are 0.5 mm apart and the interference is observed on a screen at a distance of 100 cm from the slit. It is found that the 9th bright fringe is at a distance of 7.5 mm from the second dark fringe from the centre of the fringe pattern on same side. Find the wavelength of the light used.
5. A thin glass plate of thickness t and refractive index μ is inserted between screen & one of the slits in a Young's experiment. If the intensity at the centre of the screen is I , what was the intensity at the same point prior to the introduction of the sheet.
6. In a two – slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} . If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.
7. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, calculate the positions of maxima and draw shape of interference pattern.
Take initial phase difference between the two sources to be zero.
8. A monochromatic light of $\lambda = 5000 \text{ \AA}$ is incident on two slits separated by a distance of 5×10^{-4} m. The interference pattern is seen on a screen placed at a distance of 1 m from the slits. A thin glass plate of thickness 1.5×10^{-6} m & refractive index $\mu = 1.5$ is placed between one of the slits & the screen. Find the intensity at the centre of the screen, if the intensity there is I_0 in the absence of the plate. Also find the lateral shift of the central maximum.



9. A long narrow horizontal slit lies 1mm above a plane mirror. The interference pattern produced by the slit and its image is viewed on a screen distant 1m from the slit. The wavelength of light is 600nm. Find the distance of first maximum above the mirror.
10. In Lloyd's mirror experiment, the source slit S and its virtual image S' lie in a plane 20 cm to the left from the left edge of the mirror. The mirror is 30 cm long and a screen is placed touching the right edge of the mirror. The distance of S from plane of mirror is 2 mm and wavelength of light used = 6400 Å. Find the number of maxima formed.
11. In a biprism experiment using sodium light $\lambda = 6000 \text{ Å}$ an interference pattern is obtained in which 20 fringes occupy 2 cm. On replacing sodium light by another source of wavelength λ_2 without making any other change 30 fringes occupy 2.7 cm on the screen. What is the value of λ_2 ?
12. A narrow monochromatic beam of light of intensity I is incident on a glass plate as shown in figure. Another identical glass plate is kept close to the first one & parallel to it. Each glass plate reflects 25 % of the light incident on it & transmits the remaining. Find the ratio of the minimum & the maximum intensities in the interference pattern formed by the two beams obtained after one reflection at each plate.
13. A lens ($\mu = 1.5$) is coated with a thin film of refractive index 1.2 in order to reduce the reflection from its surface at $\lambda = 4800 \text{ Å}$. Find the minimum thickness of the film which will minimize the intensity of the reflected light.
14. In Young's experiment, the source is red light of wavelength $7 \times 10^{-7} \text{ m}$. When a thin glass plate of refractive index 1.5 at this wavelength is put in the path of one of the interfering beams, the central bright fringe shifts by 10^{-3} m to the position previously occupied by the 5th bright fringe. Find the thickness of the plate. When the source is now changed to green light of wavelength $5 \times 10^{-7} \text{ m}$, the central fringe shifts to a position initially occupied by the 6th bright fringe due to red light. Find the refractive index of glass for the green light. Also estimate the change in fringe width due to the change in wavelength.
15. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength 5400 Å. It is found that the point P on the screen where the central maximum ($n = 0$) fell before the glass plates were inserted now has 3/4 the original intensity. It is further observed that what used to be the 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of the glass plate. (Absorption of light by glass plate may be neglected).



EXERCISE-III

AIEEE Previous year question

- In a double-slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}$ th of a wavelength. The ratio of the intensity of light at that point of that at the centre of a bright fringe is ; [IIT Main-2020]
(A) 0.672 (B) 0.760 (C) 0.568 (D) 0.853
- In a Young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm) _____. [IIT Main-2020]
- In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is : [IIT Main-2020]
(A) 3.9 mm (B) 6.9 mm (C) 5.9 mm (D) 4.9 mm
- In a Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be : [IIT Main-2020]
(A) 24 (B) 30 (C) 18 (D) 28
- In a Young's double slit experiment, light of 500 nm is used to produce an interference pattern. When the distance between the slits is 0.05 mm, the angular width (in degree) of the fringes formed on the distance screen is close to : [IIT Main-2020]
(A) 0.17° (B) 0.57° (C) 1.7° (D) 0.07°
- Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2 . After this the phase difference between the two waves is : [IIT Main-2020]
(A) $\frac{2\pi}{\lambda} \left(\frac{L_2}{n_1} - \frac{L_1}{n_2} \right)$ (B) $\frac{2\pi}{\lambda} \left(\frac{L_1}{n_1} - \frac{L_2}{n_2} \right)$ (C) $\frac{2\pi}{\lambda} (n_1 L_1 - n_2 L_2)$ (D) $\frac{2\pi}{\lambda} (n_2 L_1 - n_1 L_2)$
- In the figure below, P and Q two equally intense coherent sources emitting radiation of wavelength 20 m. The separation between P and Q is 5 m and the phase of P is ahead of that of Q by 90° . A, B and C are three distinct point of observation, each equidistant from the midpoint of PQ. The intensities of radiation at A, B, C will be in the ratio ; [IIT Main-2020]

(A) 0 : 1 : 4 (B) 2 : 1 : 0 (C) 0 : 1 : 2 (D) 4 : 1 : 0
- Interference fringes are observed on a screen by illuminating two thin slits 1 mm apart with a light source ($\lambda = 632.8$ nm). The distance between the screen and the slits is 100 cm. If a bright fringe is observed on a screen at a distance of 1.27 mm from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close to : [IIT Main-2020]
(A) 2 nm (B) 2.87 nm (C) $1.27 \mu\text{m}$ (D) $2.05 \mu\text{m}$



9. A Young's double-slit experiment is performed using monochromatic light of wavelength λ . The intensity of light at a point on the screen, where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. Then value of n is
- [IIT Main-2020]**
10. The light waves from two coherent sources have same intensity $I_1 = I_2 = I_0$. In interference pattern the intensity of light at minima is zero. What will be the intensity of light at maxima? **[IIT Main-2021]**
 (A) $5 I_0$ (B) $2 I_0$ (C) $4 I_0$ (D) I_0
11. A source of light is placed in front of a screen. Intensity of light on the screen is I. Two polaroids P_1 and P_2 are so placed in between the source of light and screen that the intensity of light on screen is $I/2$. P_2 should be rotated by an angle of _____ (degrees) so that the intensity of light on the screen becomes $\frac{3I}{8}$. **[IIT Main-2021]**
12. White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. The difference in wavelengths of red and violet light is _____ nm. **[IIT Main-2021]**
13. In a Young's double slit experiment, the slits are separated by 0.3 mm and the screen is 1.5 m away from the plane of slits. Distance between fourth bright fringes on both sides of central bright fringe is 2.4 cm. The frequency of light used is _____ $\times 10^{14}$ Hz. **[IIT Main-2021]**
14. The width of one of the two slits in a Young's double slit experiment is three times the other slit. If the amplitude of the light coming from a slit is proportional to the slit-width, the ratio of minimum to maximum intensity in the interference pattern is $x : 4$ where x is _____. **[IIT Main-2021]**
15. If the source of light used in a Young's double slit experiment is changed from red to violet : **[IIT Main-2021]**
 (A) The central bright fringe will become a dark fringe.
 (B) The intensity of minima will increase
 (C) consecutive fringe lines will come closer.
 (D) the fringes will become brighter.
16. In a Young's double slit experiment two slits are separated by 2 mm and the screen is placed one meter away. When a light of wavelength 500 nm is used, the fringe separation will be : **[IIT Main-2021]**
 (A) 0.25 mm (B) 0.50 mm (C) 1 mm (D) 0.75 mm
17. In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern. **[IIT Main-2021]**
 (A) 3 : 1 (B) 1 : 4 (C) 2 : 1 (D) 4 : 1
18. Two coherent light sources having intensity in the ratio $2x$ produce an interference pattern. The ratio $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be : **[IIT Main-2021]**
 (A) $\frac{\sqrt{2x}}{x+1}$ (B) $\frac{2\sqrt{2x}}{x+1}$ (C) $\frac{2\sqrt{2x}}{2x+1}$ (D) $\frac{\sqrt{2x}}{2x+1}$

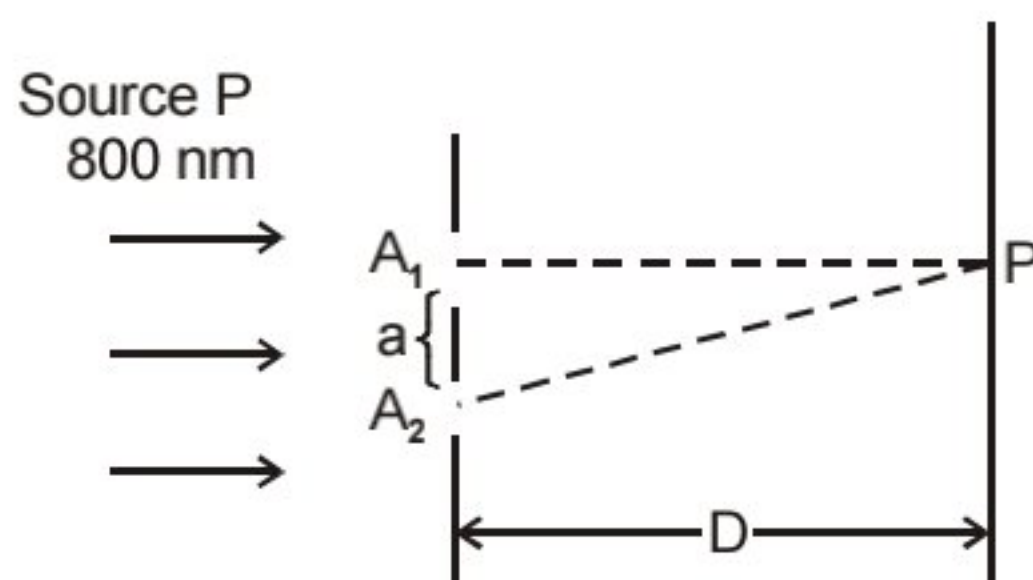


19. In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin \omega t$; where d_0 , ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as : **[IIT Main-2021]**
- (A) $\frac{\lambda D}{d_0 + a_0}$ (B) $\frac{\lambda D}{d_0^2} a_0$ (C) $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$ (D) $\frac{2\lambda D (d_0)}{(d_0^2 - a_0^2)}$
20. In Young's double slit experiment if the source of light changes from orange to blue then : **[IIT Main-2021]**
- (A) The central bright fringe will become a dark fringe.
 (B) The distance between consecutive fringes will decrease.
 (C) The distance between consecutive fringes will increase.
 (D) The intensity of the minima will increase.
21. The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is _____ mm.
 [Refractive index of air = 1.0003, wavelength of yellow light in vacuum = 6000 Å] **[IIT Main-2021]**
22. In Young's double slit arrangement, slits are separated by a gap of 0.5 mm, and the screen is placed at a distance of 0.5 m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 Å is : **[IIT Main-2021]**
- (A) 1178×10^{-9} m (B) 1178×10^{-6} m
 (C) 1178×10^{-12} m (D) 1178×10^{-7} m
23. Red light differs from blue light as they have : a **[IIT Main-2021]**
- (A) Different frequencies and different wavelengths
 (B) Different frequencies and same wavelengths
 (C) Same frequencies and same wavelengths
 (D) Same frequencies and different wavelengths
24. A fringe width of 6 mm was produced for two slits separated by 1 mm apart. The screen is placed 10 m away. The wavelength of light used is 'x' nm. **[IIT Main-2021]**
 The value of 'x' to the nearest integer is
25. In Young's double slit experiment, the fringe width is 12 mm. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm) **[JEE Main Online-2022]**
- (A) 16 (B) 9 (C) 48 (D) 12
26. Two coherent sources of light interfere. The intensity ratio of two sources is 1 : 4. For this interference pattern if the value of $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is equal to $\frac{2\alpha + 1}{\beta + 3}$, then $\frac{\alpha}{\beta}$ will be **[JEE Main Online-2022]**
- (A) 1.5 (B) 2 (C) 0.5 (D) 1
27. Two light beams of intensities 4I and 9I interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π . The difference of resultant intensities, at the point A and B, will be _____ I. **[JEE Main Online-2022]**



28. In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm. **[JEE Main Online-2022]**
29. Two beams of light having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The phase difference between the two beams are $\pi/2$ and $\pi/3$ at points A and B respectively. The difference between the resultant intensities at the two points is xI . The value of x will be _____. **[JEE Main Online-2022]**
30. Using Young's double slit experiment, a monochromatic light of wavelength 5000 Å produces fringes of fringe width 0.5 mm. If another monochromatic light of wavelength 6000 Å is used and the separation between the slits is doubled, then the new fringe width will be : **[JEE Main Online-2022]**
 (A) 0.5 mm (B) 1.0 mm (C) 0.6 mm (D) 0.3 mm
31. Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The ratio of the intensity of maxima and minima will be : **[JEE Main Online-2022]**
 (A) 2 : 3 (B) 16 : 81 (C) 25 : 169 (D) 25 : 1
32. The two light beams having intensities I and $9I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be **[JEE Main Online-2022]**
 (A) $2I$ (B) $6I$ (C) $5I$ (D) $7I$
33. The interference pattern is obtained with two coherent light sources of intensity ratio 4 : 1. And the ratio $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to **[JEE Main Online-2022]**
 (A) 3 (B) 4 (C) 2 (D) 1
34. In young's double slit experiment performed using a monochromatic light of wavelength λ , when a glass plate ($\mu \times 1.5$) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be : **[JEE Main Online-2022]**
 (A) 3 (B) 2 (C) 1.5 (D) 0.5
35. In a Young's double slit experiment, an angular width of the fringe is 0.35° on a screen placed at 2 m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index $7/5$, is $\frac{1}{\alpha}$. The value of α is _____. **[JEE Main Online-2022]**
36. In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____ nm. **[JEE Main Online-2022]**
37. In Young's double slit experiment the two slits are 0.6 mm distance apart. Interference pattern is observed on a screen at a distance 80 cm from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be _____ nm. **[JEE Main Online-2022]**

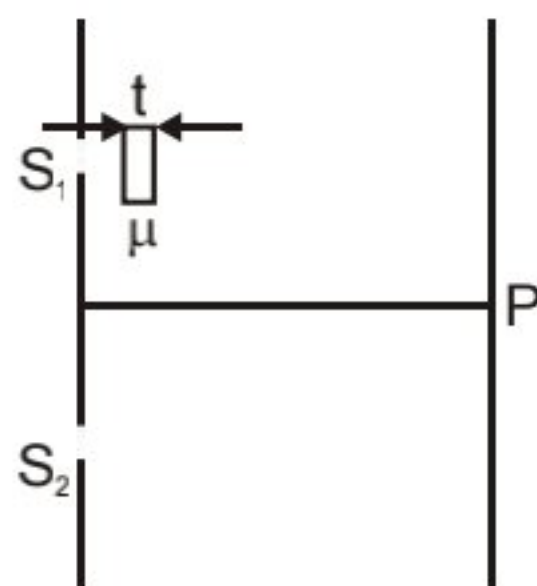
38. Given below are two statement : one is labelled as Assertion A and the other is labelled as Reason R
Assertion A : The phase difference of two light waves change if they travel through different media having same thickness, but different indices of refraction.
Reason R : The wavelengths of waves are different in different media. In the light of the above statements, choose the most appropriate answer from the options given below
 (A) Both A and R are correct but R is NOT the correct explanation of A
 (B) A is correct but R is not correct [JEE Main Online-2023]
 (C) A is not correct but R is correct
 (D) Both A and R are correct and R is the correct explanation of A
39. In a Young's double slits experiment, the ratio of amplitude of light coming from slits is 2 : 1. The ratio of the maximum to minimum intensity in the interference pattern is : [JEE Main Online-2023]
 (A) 9 : 4 (B) 9 : 1 (C) 2 : 1 (D) 25 : 9
40. The width of fringe is 2 mm on the screen in a double slits experiment for the light of wavelength of 400 nm. The width of the fringe for the light of wavelength 600 nm will be : [JEE Main Online-2023]
 (A) 3 mm (B) 4 mm (C) 1.33 mm (D) 2 mm
41. The ratio of intensities at two points P and Q on the screen in a Young's double slit experiment where phase difference between two waves of same amplitude are $\frac{\pi}{3}$ and $\frac{\pi}{2}$ respectively are [JEE Main Online-2023]
 (A) 3 : 1 (B) 1 : 3 (C) 3 : 2 (D) 2 : 3
42. A beam of light consisting of two wavelengths 7000 and 5500 Å is used to obtain interference pattern in Young's double slit experiment. The distance between the slits is 2.5 mm and the distance between the plane of slits and the screen is 150 cm. The least distance from the central fringe, where the bright fringes due to both the wavelengths coincide, is $n \times 10^{-5}$ m. The value of n is _____. [JEE Main Online-2023]
43. In a Young's double slite experiment, two slits are illuminated with a light of wavelength 800 nm. The line joining A_1P is perpendicular to A_1A_2 as shown in the figure. If the first minimum is detected at P, the value of slits separation 'a' will be : [JEE Main Online-2023]



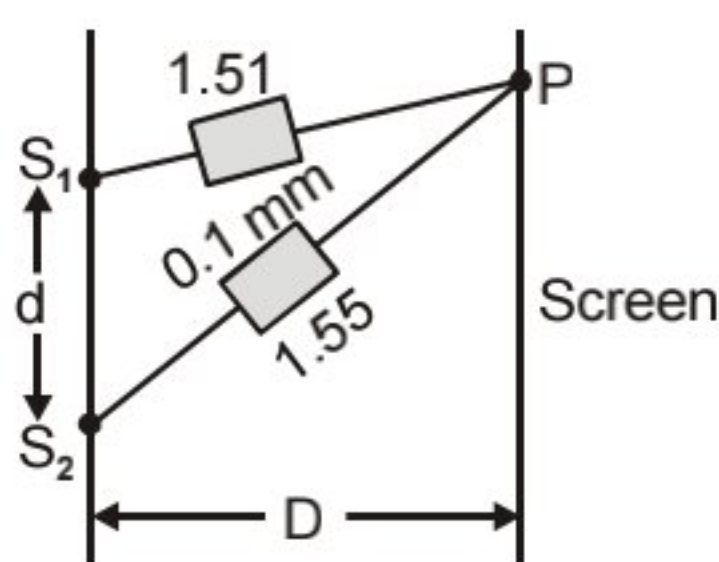
The distance of screen from slits $D = 5$ cm

- (A) 0.2 mm (B) 0.1 mm (C) 0.5 mm (D) 0.4 mm
44. In Young's double slits experiment, the position of 5th bright fringe from the central maximum is 5 cm. The distance between slits and screen is 1 m and wavelength of used monochromatic light is 600 nm. The separation between the slits is [JEE Main Online-2023]
 (A) 60 μm (B) 12 μm (C) 36 μm (D) 48 μm

45. As shown in the figure, in Young's double slit experiment, a thin plate of thickness $t = 10\mu\text{m}$ and refractive index $\mu = 1.2$ is inserted in front of slit S_1 . The experiment is conducted in air ($\mu = 1$) and uses a monochromatic light of wavelength $\lambda = 500\text{ nm}$. Due to the insertion of the plate, central maxima is shifted by a distance of $x\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is _____. [JEE Main Online-2023]



46. In Young's double slit experiment, two slits S_1 and S_2 are 'd' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness 0.1 mm but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000\text{ Å}$) from S_1 and S_2 respectively. The central bright fringe spot will shift by _____ number of fringes. [JEE Main Online-2023]



47. In a Young's double slit experiment, the intensities at two points, for the path differences $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1 + I_2}{I_0} = \underline{\hspace{2cm}}$. [JEE Main Online-2023]

48. Two light waves of wavelengths 800 and 600 nm are used in Young's double slit experiment to obtain interference fringes on a screen placed 7 m away from plane of slits. If the two slits are separated by 0.35 mm, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be _____ mm. [JEE Main Online-2023]

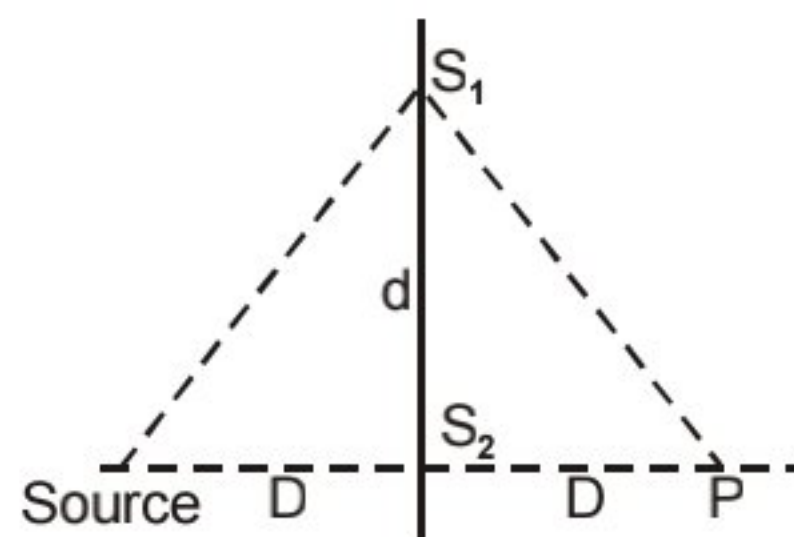
49. In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{7\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is [JEE Main Online-2024]

- (A) $\frac{3}{4}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

50. Two waves of intensity ratio 1 : 9 cross each other at a point. The resultant intensities at that point, when (a) Waves are incoherent is I (b) waves are coherent is I_2 and differ in phase by 60° . If $\frac{I_1}{I_2} = \frac{10}{x}$ then $x = \underline{\hspace{2cm}}$. [JEE Main Online-2024]

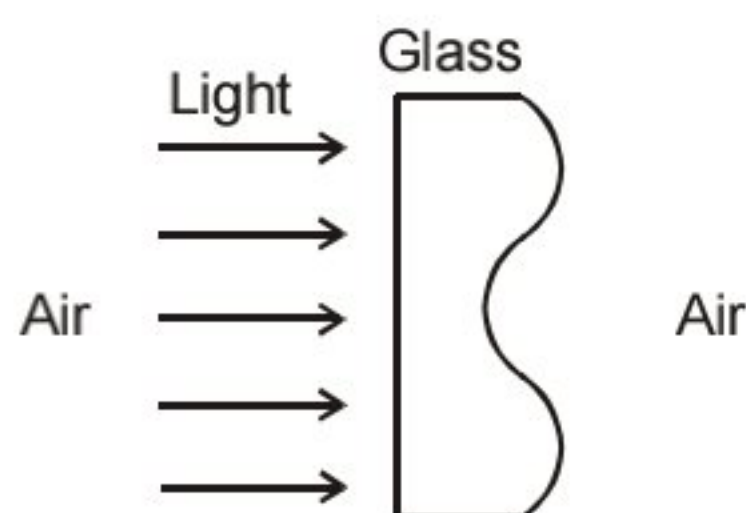


51. In Young's double slit experiment, monochromatic light of wavelength $5000 \text{ \AA}$ is used. The slits are 1.0 mm apart and screen is placed at 1.0 m away from slits. The distance from the centre of the screen where intensity becomes half of the maximum intensity for the first time is $\text{_____} \times 10^{-6} \text{ m}$. [JEE Main Online-2024]
52. In a double slit experiment shown in figure, when light of wavelength 400 nm is used, dark fringe is observed at P. If $D = 0.2 \text{ m}$, the minimum distance between the slits S_1 and S_2 is _____ mm . [JEE Main Online-2024]



EXERCISE-IV

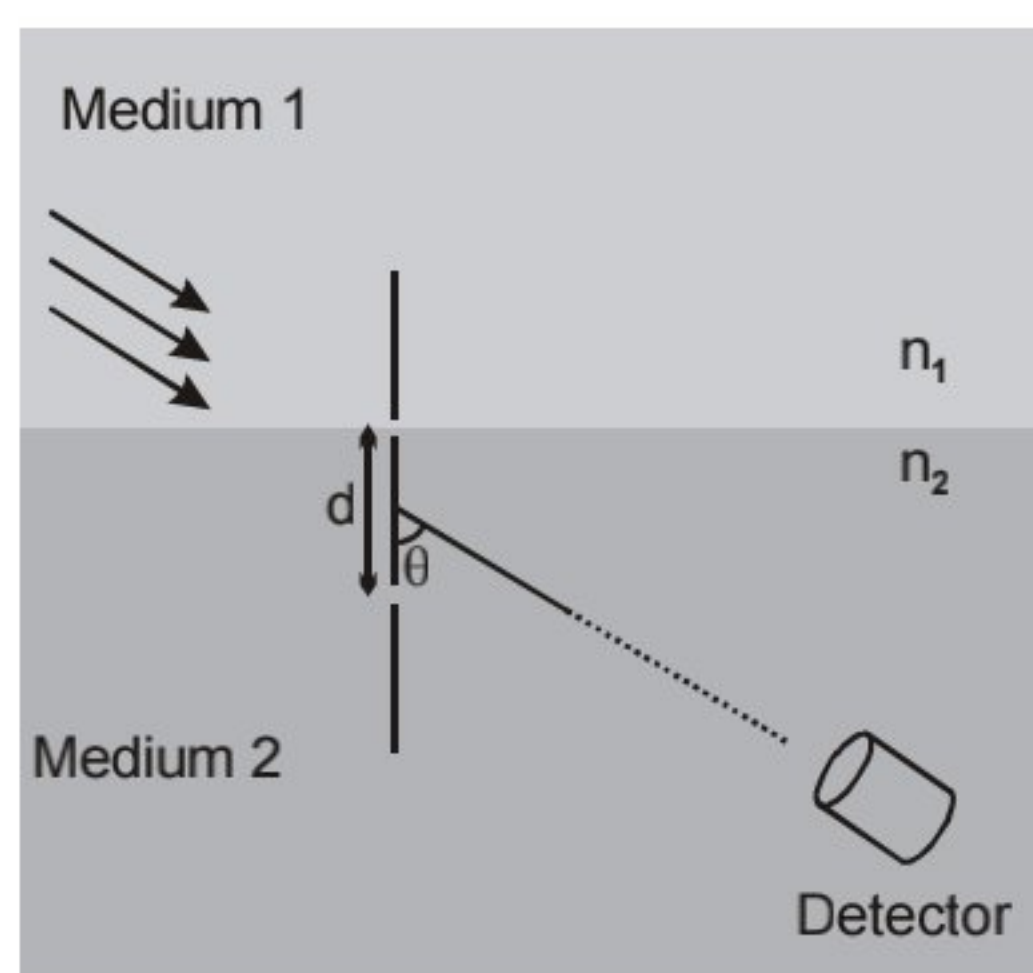
1. A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wavefront will be (figures are schematic and not drawn to scale)



[JEE Advanced-2020]

2. A double slit setup is shown in the figure. One of the slits is in medium 2 of refractive index n_2 . The other slit is at the interface of this medium with another medium 1 of refractive index $n_1 (\neq n_2)$. The line joining the slits is perpendicular to the interface and the distance between the slits is d . The slit widths are much smaller than d . A monochromatic parallel beam of light is incident on the slits from medium 1. A detector is placed in medium 2 at a large distance from the slits, and at an angle θ from the line joining them, so that θ equals the angle of refraction of the beam. Consider two approximately parallel rays from the slits received by the detector.

[JEE Advanced-2022]

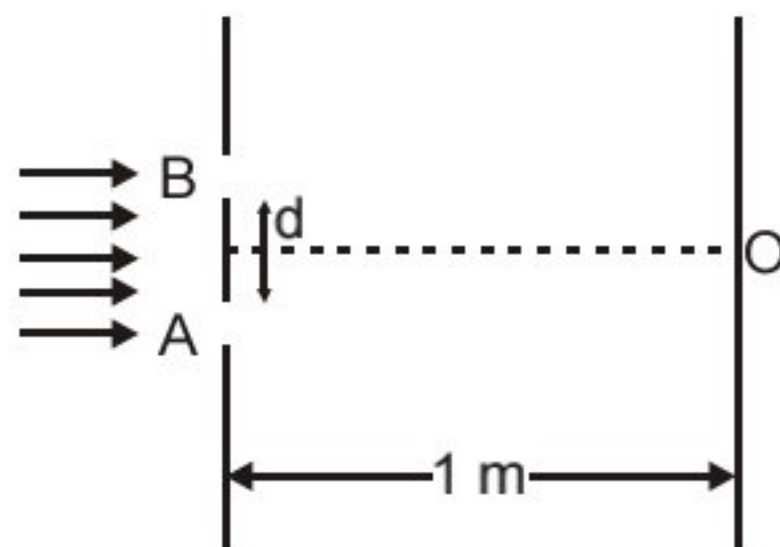


Which of the following statement(s) is(are) correct?

- (A) The phase difference between the two rays is independent of d
- (B) The two rays interfere constructively at the detector
- (C) The phase difference between the two rays depends on n_1 but is independent of n_2
- (D) The phase difference between the two rays vanishes only for certain values of d and the angle of incidence of the beam, with θ being the corresponding angle of refraction.

Paragraph (Q.3 to Q.4)**[JEE Advanced-2024]**

In a Young's double slit experiment, each of the two slits A and B, as shown in the figure, are oscillating about their fixed center and with a mean separation of 0.8 mm. The distance between the slits at time t is given by $d = (0.8 + 0.04 \sin \omega t)$ mm, where $\omega = 0.08 \text{ rad s}^{-1}$. The distance of the screen from the slits is 1 m and the wavelength of the light used to illuminate the slits is $6000 \text{ \AA}$. The interference pattern on the screen changes with time, while the central bright fringe (zeroth fringe) remains fixed at point O.



3. The 8th bright fringe above the point O oscillates with time between two extreme positions. The separation between these two extreme positions, in micrometer (μm), is _____
4. The maximum speed in $\mu\text{ m/s}$ at which the 8th bright fringe will move is _____.

ANSWER KEY

EXERCISE-I

1. C 2. C 3. C 4. A 5. A 6. B 7. D 8. C 9. D
 10. B 11. C 12. C 13. C 14. C 15. D 16. A 17. D 18. A
 19. D 20. C 21. B 22. B 23. A 24. A 25. A 26. B 27. A
 28. A 29. A 30. A
 31. [(A)–P, (B)–S, (C)–Q, (D)–R 32. (A) P,Q; (B) P,Q,R,S; (C) none; (D) P

EXERCISE-II

1. (ii) medium-I 2. 2 3. 5000 Å 4. 5000 Å
 5. $I_0 = I \sec^2 \left[\frac{\pi(\mu - 1)t}{\lambda} \right]$ 6. 6000 Å 7. 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.; 
 8. 0, 1.5 mm 9. 0.15 mm 10. 38 11. 5400 Å 12. 1 : 49
 13. 10^{-7} m 14. $7 \mu\text{m}$, 1.6, $\frac{400}{7} \mu\text{m}$ (decrease) 15. $9.3 \mu\text{m}$

EXERCISE-III

1. D 2. 750 3. C 4. D 5. B 6. C 7. B
 8. C 9. 9.00 10. C 11. 30.00 12. 300.00 13. 5.00 14. 1.00
 15. C 16. A 17. D 18. C 19. C 20. B 21. 2.00
 22. B 23. A 24. 600.00 25. B 26. B 27. 24 28. 630
 29. 2 30. D 31. D 32. B 33. B 34. B 35. 4.00
 36. 600 37. 450 38. D 39. B 40. A 41. C 42. 462
 43. A 44. A 45. 4 46. 10 47. 3 48. 48 49. D
 50. 13.00 51. 125.00 52. 0.20

EXERCISE-IV

1. A 2. A, B 3. 598.00 TO 502.00 4. 23.50 TO 24.50